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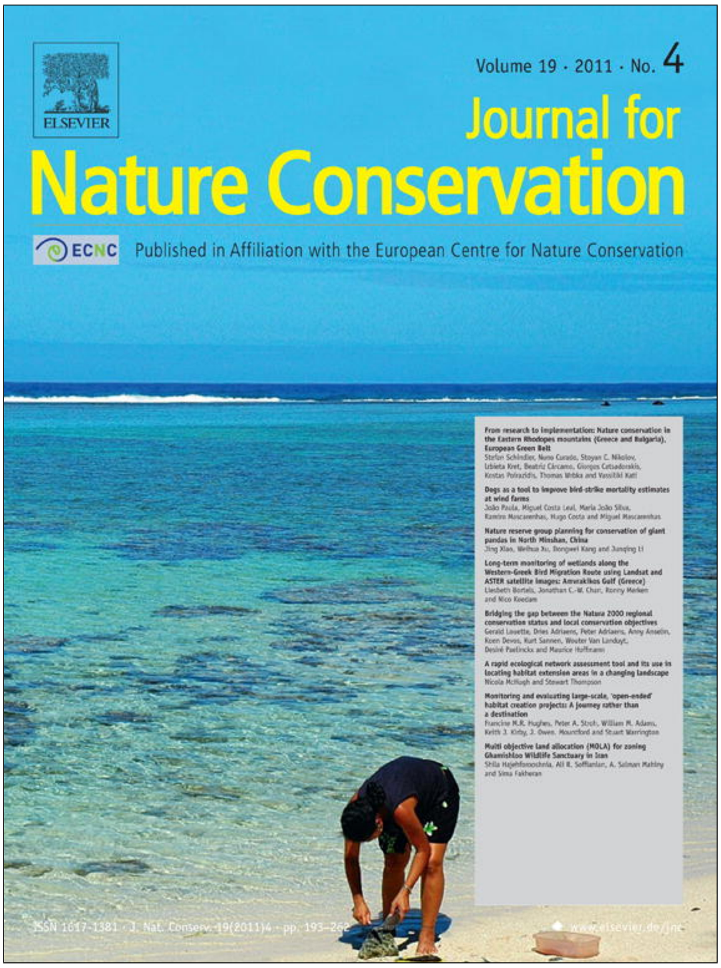

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From research to implementation: Nature conservation in the Eastern Rhodopes mountains (Greece and Bulgaria), European Green Belt

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ABSTRACT

Nature conservation should ideally build on the scientific recommendations that are concluded from applied conservation research, as well as on monitoring schemes that evaluate the effectiveness of recommendations. We considered as a case study a system of six protected areas located in the Eastern Rhodopes mountains in the southern part of the European Green Belt (EGB). To investigate nature conservation effectiveness, we reviewed 196 articles from scientific journals and books, eight doctoral and master theses, and 39 scientific reports regarding the Greek (one protected area, 428 km^2) and the Bulgarian (five protected areas, 904 km^2) part of the study area. We extracted 743 conservation recommendations, and through questionnaires completed by 10 local experts, we found that 74% of the recommendations were familiar for the experts. In the Greek (GR) and the Bulgarian part (BG) only 52% and 16%, respectively, of the recommendations were implemented, and only 15% (GR) and 3.1% (BG) were implemented *and* evaluated regarding their effectiveness. According to the experts, the main reasons for non-implementation and non-evaluation were absence or incompetence of the responsible authorities. Some recommendations obtained a remarkable low rate of implementation, such as those regarding agriculture and livestock rearing practices (GR: 29%, BG: 16%) or mammal conservation (GR: 0%, BG: 16%). Some other recommendations obtained rather high rates at least for Greece, such as tourism and environmental education (GR: 57%, BG: 42%) and bird conservation (GR: 57%, BG: 11%). We found that researchers and conservation managers at both sides of the Greek-Bulgarian border face similar implementation problems, related often to the lack of political will for nature conservation and establishment of competent authorities. The role of the EGB is crucial in enhancing the established cross-border collaborations between stakeholders involved in nature conservation.

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Introduction

Although nature conservation activities have increased substantially over the last decades and environmental change and biodiversity conservation are currently highly ranked in the political agendas worldwide (Pullin & Knight, 2001, 2009), it is widely recognised that the 2010 target of halting the loss of biodiversity was not achieved (EEA, 2007; Fisher, 2009). Considerable time and financial resources have been spent in addressing conservation recommendations all over the world, but it is still a common situation that only few of them have been efficiently implemented (Mauerhofer, 2010). Moreover, a large amount of conservation efforts have never been evaluated or monitored, leaving a gap in our knowledge about whether the conservation objectives have been achieved (Pullin & Knight, 2001, 2005). Therefore, a recognised challenge in the post 2010 era is to directly link scientific knowledge with policy-making and *in situ* effective implementation of conservation actions (EPBRS, 2009; Pressey & Bottrill, 2009; Pullin et al., 2009).

Such conservation actions and policies are focused mainly in networks of protected areas, given that they constitute the cornerstone of global conservation effort (Chape, Harrison, Spalding, & Lysenko, 2005; Jones-Walters, 2007; Jongman, 1995; Jongman, Külvik, & Kristiansen, 2004). The literature abounds of evidence about the importance and inadequacies of protected areas and

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other conservation strategies (Chape et al., 2005; Gaston, Jackson, Cantú-Salazar, & Cruz-Piñón, 2008; Gaston, Jackson, Nagy, Cantú-Salazar, & Johnson, 2008; Mas, 2005; Nikolov, 2010; Wrbka, Schindler, Pollheimer, Schmitzberger, & Peterseil, 2008). However, even within ecological networks, conservation targets cannot be achieved without assessments on whether the proposed conservation recommendations have been implemented and once implemented, if they operate effectively (Pullin & Knight, 2005). In this study we attempted for the first time to evaluate in a systematic way the real implementation value of scientific conservation recommendations, taking as a case study the Eastern Rhodopes mountains, in the Greek-Bulgarian part of the European Green Belt (EGB).

The EGB, running 12500 km throughout Europe along the former Iron Curtain is the backbone of an ecological network, and a living monument and global symbol for transboundary cooperation in nature conservation and sustainable development (Terry, Ullrich, & Riecken, 2006). Today this string of protected habitats connects European landscapes in which intensive economic development was inhibited from 1945 until 1989. The Eastern Rhodopes are an important element of the EGB and other national and international ecological networks, i.e. IBAs, Natura 2000 (Kostadinova & Gramatikov, 2007; Stoychev & Petrova, 2003), and maintain a particularly high level of biodiversity (Beron & Popov, 2004; Catsadorakis & Källander, 2010; Griffiths, Krytufek, & Harrison, 2004; Myers, Mittermeier, Mittermeier, da Fonseca, & Kent, 2000; Temple & Terry, 2007). The area has been comparatively well studied (Beron & Popov, 2004; Catsadorakis & Källander, 2010) and it should be expected that many conservation recommendations have been developed. Administrative isolation has led to no integrated management of protected areas to date from both sites of the border, although such recommendation exists (Beron & Popov, 2004; Vasilakis, Poirazidis, & Elorriaga, 2008). Since Bulgaria recently joined the EU, it is crucial to analyse and compare management strategies between Bulgaria and Greece to increase the effectiveness and integration of the existing ecological networks.

The opinion of both Greek and Bulgarian governmental authorities on the status of protected areas is positive, but proposed policy measures (including management measures and regulations) are rarely implemented in practice (Apostolopoulou & Pantis, 2009; Dimitrakopoulos, Memtsas, & Troumbis, 2004; Liarikos, 2006; Mateeva, 2009; Papageorgiou & Vogiatzakis, 2006). Although NGOs and environmental scientific organisations have put a lot of effort into lobbying for the correct implementation of the elaborated conservation recommendations (e.g. Catsadorakis, 2010; Catsadorakis et al., 2010; Kostadinova & Gramatikov, 2007; Mateeva, 2009), a systematic analysis about effectiveness and flaws of nature conservation measures has not been undertaken so far in any of the two countries. The main objectives of the present study were to summarise conservation recommendations for several model sites in the Eastern Rhodopes mountains, and to analyse for the Greek and the Bulgarian part the level of implementation and the reasons for their non-implementation. Further aims were to assess the degree of evaluation of effectiveness, the reasons for non-evaluation, and the sources of information uptake of local conservation managers.

Methods

Study area

The Eastern Rhodopes mountains (Fig. 1) occupy about 6000 km^2 shared between Greece (1800 km^2) and Bulgaria (4200 km^2) (Beron & Popov, 2004). They are characterised by Continental-Mediterranean climate and a hilly and low mountainous landscape with altitudes ranging from 0 to 1483 m (Beron & Popov, 2004). Specific natural and cultural values occur commonly in both countries, including traditional pastoralism and the resulting landscape heterogeneity, old-growth pine and oak forests and a few other rare habitats, high level of endemic and rare plant and animal species, high diversity and population density of raptorial birds, and finally a variety of geological and cultural monuments (Beron & Popov, 2004; Catsadorakis & Källander, 2010; Poirazidis, Skartsi, & Catsadorakis, 2002).

In the Greek part of Eastern Rhodopes we limited our study to one large protected area, Dadia-Lefkimi-Soufli Forest National Park (Dadia NP), covering 428 km^2, including two strictly protected core areas (78 km^2). We did not consider the three established protected areas Treis Vryses (99 km^2), Oreinos Evros (489 km^2) and Potamos Filouris (21 km^2), (Schlumprecht & Ludwig, 2007), as we were not aware of any publications regarding these areas. Dadia NP is a hilly area (altitudes ranging from 20 to 645 m), covered by extensive pine and oak forest and characterised by a heterogeneous landscape (Kati et al., 2010; Poirazidis, Goutner, Skartsi, & Stamou, 2004; Schindler, Poirazidis, & Wrbka, 2008; Schindler et al., 2010). It is an essential refuge for breeding populations of a unique assemblage of raptors (Poirazidis, Schindler, Ruiz, & Scandolara, 2011; Poirazidis et al., 2011; Schindler, 2010), containing the only remaining black vulture (*Aegypius monachus*) breeding colony in the Balkan Peninsula (Poirazidis, Goutner, Skartsi, & Stamou, 2004; Skartsi et al., 2008), and a high diversity of passerines (Kati & Sekercioglou, 2006), amphibians and reptiles (Kati et al., 2007), butterflies (Grill & Cleary, 2003), grasshoppers (Kati, Dufrêne, Legakis, Grill, & Lebrun, 2004), and vascular plants (Kati, Lebrun, Devillers, & Papaioannou, 2000; Korakis, Gerasimidis, Poirazidis, & Kati, 2006). The management and conservation of forested lands in Greece (including National Parks) is administered by the Greek Forest Service, while the Ministry of the Environment (represented by the regional governmental units) has the responsibility for the implementation of conservation measures in non-forested areas. Additionally, in some national parks (such as Dadia NP where such an agency was founded in 2003), a management agency has been established to coordinate all the conservation and monitoring activities as well as to promote eco-tourism initiatives.

The Bulgarian part of Eastern Rhodopes includes five special protected areas (SPAs) totalling 904 km^2 (Fig. 1): Arda Bridge (150 km^2); Byala Reka (446 km^2); Krumovitza (112 km^2); Madzharovo (36 km^2); and Studen Kladenets (160 km^2). The whole area is characterised by exceptional biodiversity, including about 50% of Bulgarian flora, 70% of Bulgarian herpetofauna, and 70% of Bulgarian avifauna (Beron & Popov, 2004). Furthermore, the National Strategy for Conservation of Biodiversity (NSCB) considers the Bulgarian part of Eastern Rhodopes as a priority area for the creation of new protected areas at a national scale, because of its great importance concerning species diversity, endemism and rarity. The study areas in Bulgaria do not have their own administration, as they are not National or Natural Parks. Instead, nature conservation is regulated by (1) the Ministry of Environment and Waters (MOEW), applied at the regional level by the Regional Inspection of Environment and Waters, (2) the Ministry of Agriculture and Food, applied at a regional level by the State Forestry Agency and their regional sub-units, and (3) the Municipality administration authorities. Responsible for the implementation of conservation measures is the MOEW.

Literature review

To collect scientific literature, we used the SCOPUS online search machine and the following terms for title, abstract and keywords: "(Eastern Rhodopes OR southern Bulgaria OR north-eastern Greece) AND (biodiversity OR conservation)". We additionally considered the literature compiled in recent scientific books about the Eastern Rhodopes (Beron & Popov, 2004; Catsadorakis & Källander, 2010;

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Fig. 1. The Eastern Rhodopes mountains, located in northeastern Greece and southern Bulgaria, and the six protected areas, for which conservation recommendations were detected, extracted and analysed in the frame of this study.

Kostadinova & Gramatikov, 2007; Stoychev & Petrova, 2003). We reviewed the collected articles that included peer reviewed ones such as journal articles; short communications; book chapters; conference proceedings; and papers in local journals as well as not necessarily peer-reviewed ones such as unpublished reports and doctoral and master theses. We thoroughly reviewed and included in our analyses all peer reviewed literature; but only the local and grey literature that was assessed as relevant and not repetitive to scientific publications. We inventoried all conservation recommendations from the literature and extracted for each recommendation its conservation goal and the name of the specific area and taxonomic group it concerned (see Appendix D).

Evaluation of recommendation implementation

We distributed a questionnaire to five local conservation experts in each country to assess the implementation of the recommendations (Appendix A). Several questions and categories were adapted from a survey of management-plan compilers in the United Kingdom and Australia (Pullin & Knight, 2005). The experts were selected based on their great experience and overview about the conservation management in the area and had published at least one scientific paper on this issue. The recommendations were randomly and equally distributed to the experts for evaluation. The experts had to answer for each recommendation: (a) whether they ever heard about it and from which source; (b) whether the recommendation was implemented or not and why, and in case of implementation; whether (c) justification was given for the implementation; and (d) whether the effectiveness of the recommendation was evaluated.

Data analysis

In our analysis, we grouped the recommendations in the following eleven categories: legislation; administration and regulation; research; monitoring; landscape conservation; forest management; agriculture and livestock rearing; wildlife management; hunting and fishing; tourism and environmental education; and sustainable development. We assessed the level of awareness of the recommendations by the local conservation experts, the experts' sources of information uptake, and the degree of implementation of recommendations as proportion of those implemented from the total recommendations proposed for each taxonomic group and for each of the above categories (Appendix D). In the same way, we assessed the degree of evaluation per taxon and per category.

Results

Reviewed literature

We reviewed 119 and 124 articles respectively for the Greek (GR) and the Bulgarian (BG) part of the Eastern Rhodopes (Table 1). Thereof 101 articles had as their main topic nature conservation, and a total of 743 (GR: 324, BG: 419) recommendations were extracted from 105 articles (Table 1, see Appendix C for articles, and Appendix D for recommendations). Particularly in Bulgaria, there was a lack of conservation recommendations in scientific papers. In the 31 reviewed journal papers, we detected only one recommendation, while 416 recommendations were detected in book chapters and unpublished reports (Table 1).

Implementation, justification and evaluation

The local conservation experts were aware of most of the recommendations (79% GR, 70% BG): in the Greek part mainly from existing management plans (42%), in Bulgaria mainly from unpublished reports (43%) and books (33%) (Fig. 2). More than half (52%) of the Greek but only 16% of the Bulgarian recommendations were

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Table 1
Reviewed articles and extracted recommendations for the Greek (GR) and the Bulgarian (BG) part of the Eastern Rhodopes.

Source	Detected articles		Reviewed articles		With topic nature conservation		Articles with conservation recommendations		Number of conservation recommendations	
	GR	BG	GR	BG	GR	BG	GR	BG	GR	BG
Per reviewed literature										
Journal papers	70	42	55	31	20	3	29	1	99	1
Short communications	21	8	17	7	4	0	4	0	11	0
Book chapters	15	54	12	53	8	15	9	13	34	198
Conference proceedings	21	9	19	2	8	0	10	0	46	0
Papers in local journals	87	19	0	0	0	0	0	0	0	0
Ph.D./M.Sc. theses	32	3	6	2	6	1	5	1	23	2
Reports	86	31	10	29	6	30	9	24	111	218
TOTAL per country	332	166	119	124	52	49	66	39	324	419
TOTAL in GR & BG	487		243		101		105		743	

implemented. In both countries, the main reasons given for non-implementation were the lack of sufficient competence in the responsible authorities (44% GR, 17% BG), the absence of responsible authorities (28% GR, 10% BG), and costly implementation of the recommendation (23% GR, 10% BG) (Fig. 2). In Bulgaria, most of the reasons for the non-implementation of proposed recommendations were not listed in our questionnaire ("other") and were related to the "lack of political will".

Justification was given for 79% of the Greek implementations (mostly in the form of management plans, 49%) but for only 31% of the Bulgarian ones (mostly in the form of unpublished reports and books, 43% and 33%, respectively) (Fig. 3). For only a minority of recommendations not currently implemented, experts could confirm that they will be implemented soon (8% GR, 1.5% BG). A very small proportion of the implemented recommendations has been evaluated regarding their effectiveness (as proportion from all recommendations: 15% GR, 3.1% BG; as proportion from the implemented recommendations: 29% GR, 19% BG). Main reasons were the insufficient competence of the responsible authorities (GR: 46%, BG: 38%), and the absence of such authorities (GR: 29%, BG: 31%)" (Fig. 3).

Taxa and categories of recommendations

Scientific research focused mostly on birds (51 articles), plants (27) and invertebrates (13) in Greece, and on invertebrates (44), reptiles (11) and birds (10) in Bulgaria, while the least studied groups were generally amphibians and particularly fish (Appendix

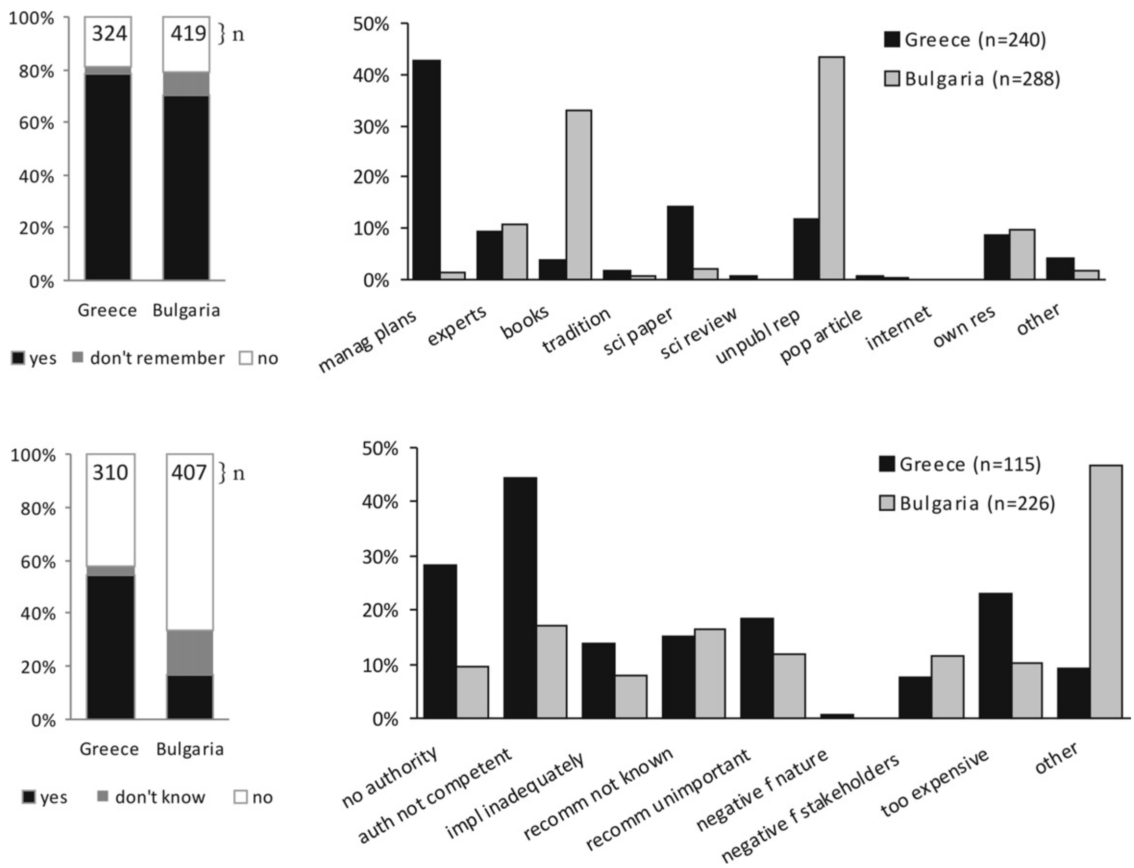

Fig. 2. Sources of information uptake by conservation experts and rates of implementation of published conservation recommendations for the Greek and the Bulgarian part of the Eastern Rhodopes Mountains. Answers to the questions "Have you ever heard before about this recommendation, and if yes, from which source?" (upper panels) and "Is this recommendation implemented in your area and if not, why not?" (lower panels). *N*-values show the number of recommendations for which an answer was provided by the experts. For the exact formulation of the chosen options, see Appendix A.

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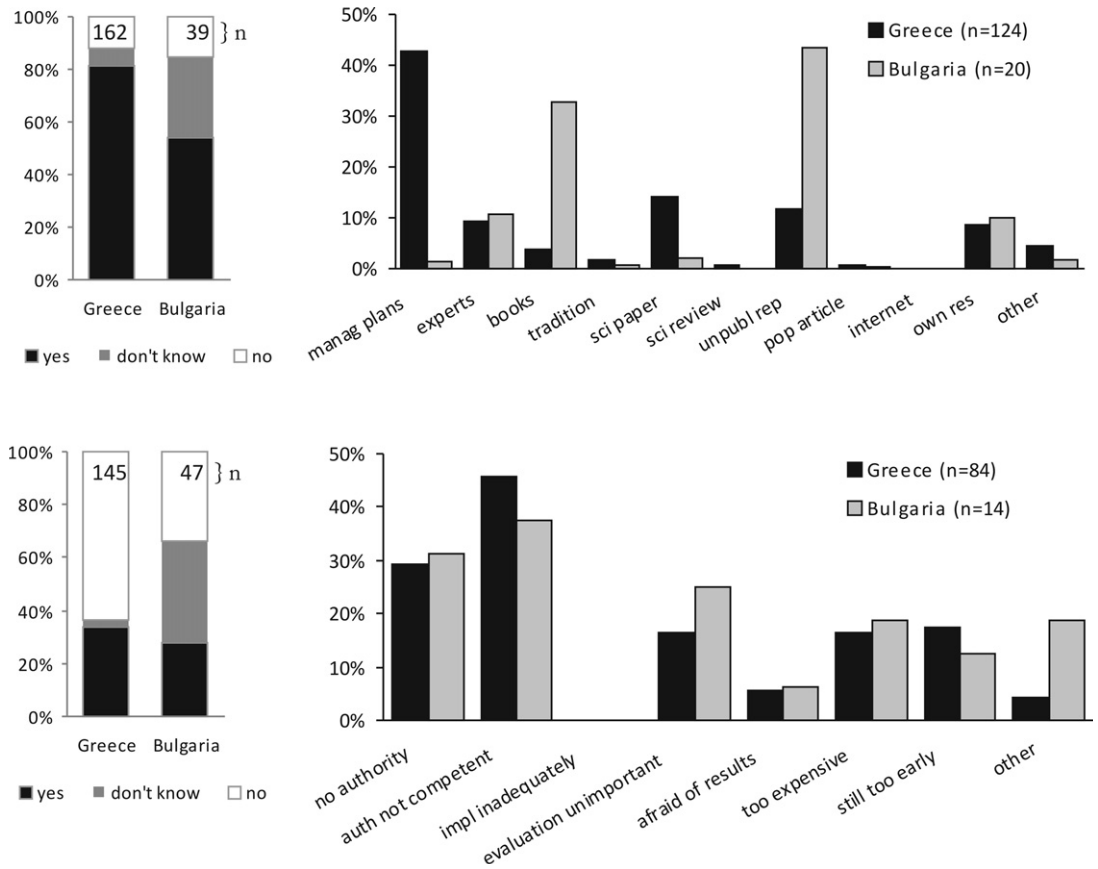

Fig. 3. Rates of justification, and degree of evaluation of implemented conservation recommendations for the Greek and the Bulgarian part of the Eastern Rhodopes Mountains. Answers to the questions "Has justification been given for its implementation, if yes in form of" (upper panels) and "If implemented, is the effectiveness of the implemented recommendation evaluated in your area, and if not, why not?" (lower panels). *N*-values show the number of recommendations for which an answer was provided by the experts. For the exact formulation of the chosen options, see Appendix A.

B). In Greece, the recommendations regarding bird conservation were implemented at a rather high rate (57%) particularly for black vulture (76%) and other raptors (47%) (Appendix B). Conversely, in Bulgaria the overall rate of implementation of recommendations concerning birds was very low (10%). On the other hand, in Bulgaria conservation recommendations were well implemented for mammals (30%), but no implementation exists in Greece for this group (Fig. 4). Few recommendations were implemented for herpetofauna and fish, for invertebrates the rates were 42% (GR) and 17% (BG), and for plants 7% (GR) and 15% (BG). Rates of evaluations were low throughout all taxa, for most of them evaluations were not performed. The highest value (16%) was obtained for birds in Greece (Fig. 4).

Most of the recommendations dealt with forest management, legislation (especially in BG), wildlife management, administration and regulation, and tourism and environmental education (Fig. 5). The highest proportions of implementation were obtained for recommendations regarding tourism and environmental education (GR: 57%, BG: 42%), administration and regulation (GR: 66%, BG: 12%), and monitoring (GR: 55%, BG: 28%), while the weakest implementation rates were obtained for recommendations regarding agriculture and live-

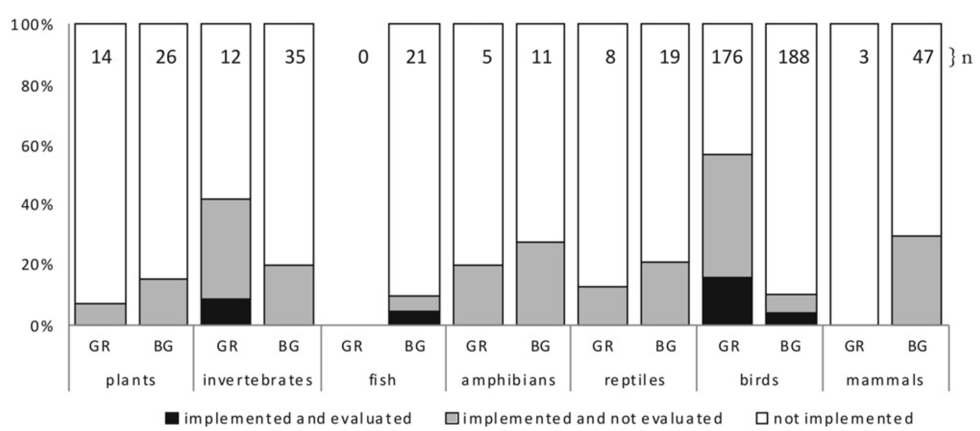

Fig. 4. Taxon specific implementation and evaluation rate of conservation recommendations in the Greek (GR) and Bulgarian (BG) part of the Eastern Rhodopes mountains. n = number of detected and considered recommendations per taxa.

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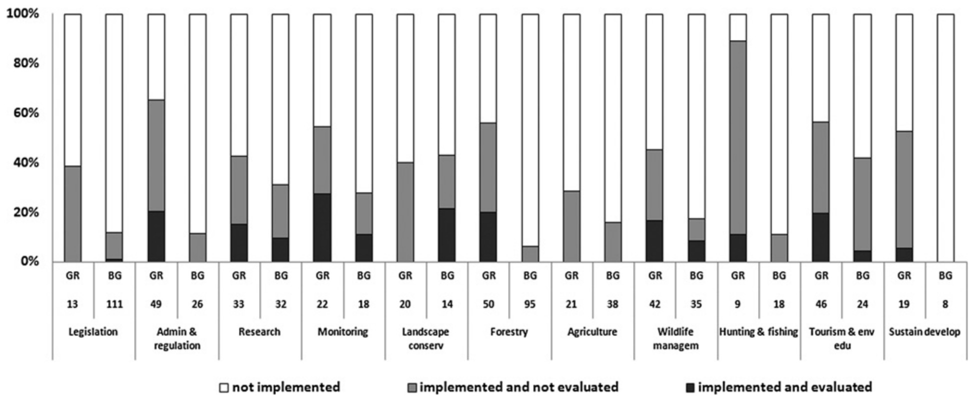

Fig. 5. Proportion of implemented and evaluated recommendations in Greece (GR) and Bulgaria (BG). The number of recommendations is presented below the country codes. The recommendations had been assigned to the categories legislation, administration, research; monitoring; landscape conservation; forest management; agriculture and livestock rearing; wildlife management; hunting and fishing; tourism and environmental education; and sustainable development.

stock rearing (GR: 29%, BG: 16%), and legislation (GR: 39%, BG: 12%).

Discussion

Constraints of evidence-based conservation

In the light of continuous biodiversity loss and global environmental change, there is an urgent need not only for further research and better understanding of our natural world, but even more for concrete synthesis and implementation of the current state of knowledge in practice. We have a scrappy knowledge of biodiversity patterns and natural processes, introducing uncertainty as an inherent characteristic to any conservation decision (Hey, Waples, Arnold, Butlin, & Harrison, 2003; Meffe & Carroll, 1994; Regan et al., 2005). Furthermore, we lack a strong mechanism to synthesise, evaluate and communicate the current state of our knowledge as concrete evidence-based recommendations for nature conservation worldwide, accessible and available to decision-makers and managers (Fisher, 2009; Loreau & Oteng-Yeboah, 2006; Pullin & Knight, 2009). Conservationists cannot benefit from predefined and universal prescriptions to solve environmental problems, differentiating between good and harmful management practices and human interventions. Taking as a case study an ecologically important area in the SE European Green-Belt, this study proved furthermore that even in cases where high-standard scientific research is available and solutions are provided through precise conservation recommendations, there is a weakness in implementing them in practice and an even greater weakness in evaluating their effectiveness. A lack of evaluations of effectiveness is particularly problematic, because chances get missed for both, essential improvements in conservation measures, and convincing arguments for decision makers in favour of urgent implementations of conservation measures (Pullin & Knight, 2005).

One of the main problems of evidence-based conservation is that primary scientific research rarely reaches local conservation managers (Pullin & Knight, 2001, 2005). This problem can be confirmed by our results. Additionally, at least for Bulgaria, scientific papers very rarely contain any conservation recommendation. Thus, even the conservation experts chosen for this study, all persons who read and publish scientific papers, rarely obtained their information from primary scientific literature.

Differences between Bulgaria and Greece

A main element enhancing conservation is the management plan (Anderson, Forcey, Osbourne, & Spurgeon, 2002). In this study, the implementation rate was more than three times higher in Greece than in Bulgaria, and also the main sources of information uptake and for justifications for implementation differed between the countries. In Greece, mainly management plans were used, while in Bulgaria management plans are only obligatory for National Parks and Natural Parks, and are often missing in other protected areas (Kostadinova & Gramatikov, 2007), and the main source were books and unpublished reports. Obviously, a good management plan is not a panacea but an important source of information for conservation managers (Pullin & Knight, 2005) and an important step towards effective nature conservation. Further, the majority of Bulgarian scientific literature sources related to biodiversity issues did not discuss any conservation problems and recommendations, while the main part of the unpublished reports (made by nature-protective NGOs) was focused on this. These facts can partly be caused by values and style of scientific writing in Bulgaria, where priority is given to the pure descriptive studies (i.e. faunistics and floristics), while nature-protective NGOs are often constrained by other priorities thus inhibiting the publication of their concepts and studies in scientific journals. This probably influenced the “quality” of the recommendations as most of the Bulgarian ones resulted from experts' knowledge, while only few of them were evidence-based.

Another difference among old and new EU member states is the stage of development of the Natura 2000 process. Although in Greece the system is far from functioning well in practice (Apostolopoulou & Pantis, 2009; WWF Greece, 2004, 2007), Bulgaria joined the EU and its mechanisms recently and is thus still limping behind in the implementation of the conservation directives. In Greece the designation of 359 Greek Sites of Community Importance (2006/613/EU) is finalised, and 27 management agencies were established in 61 Greek Natura 2000 sites (Apostolopoulou & Pantis, 2009), including Dadia NP in the Eastern Rhodopes. In Bulgaria, the draft list for the Natura 2000 network included 551 provisional sites covering about 34% of the national territory (without marine sites) (WWF, 2006). The implementation of the network was retarded by the lack of specific budget lines established for the implementation of Natura 2000 and by postponing of 26 SPAs (WWF, 2006). Currently, a total of 114 SPAs (based on the existing IBAs) and 225 SCIs are established, covering a total of 33.8% of the state territory (about 24% and 30%, respectively), but very few of them have management plans and agencies (Kostadinova & Gramatikov, 2007; WWF, 2006).

Greece and Bulgaria differ in the development of their nature conservation activities, especially in those from the public sector. The main drawback in Bulgaria seems to be that local authorities that should deal with conservation issues do not exist yet, and that

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there is little political will for creating such authorities. In Greece, management agencies do exist for a minority of protected areas, but their competence seems to be insufficient, for reasons such as unclear mandate, low degree of social acceptance and inability to employ high quality permanent scientific and administrative staff due to poor funding and other restrictions. Further reasons for the inadequacies in both countries should be sought in the lack of vertical and horizontal coordination among state services, the huge overlaps and gaps of responsibilities, the perplexed legal systems, the poor spatial planning systems and, ultimately, the almost complete absence of political commitment to conservation coupled with economic interests related to the territory of the potential protected areas, and a high level of bureaucracy (Apostolopoulou & Pantis, 2009; Liarikos, 2006; Mateeva, 2009; Papageorgiou & Vogiatzakis, 2006).

The importance of the conservation history

Pullin and Knight (2005) detected in their survey among UK and Australian management plan compilers that the differences in conservation history was a main reason for the different results between the two countries. Similarly, the conservation history differs in our case between the Greek and the Bulgarian part of the Eastern Rhodopes mountains. In Dadia National Park, Greece, raptors and vultures were identified from the beginning of its designation as a Nature Reserve as the main conservation value of the area. As this large forested area was very sparsely populated, the dedicated involvement of WWF Greece in the area allowed the achievement of the minimum necessary conditions, alliances and partnerships with the local authorities, to enable the implementation of many of the necessary conservation measures. The comparatively high implementation and evaluation rates should be attributed to the catalytic, continuous long term presence of the environmental NGO WWF Greece, which had a rightly focused conservation strategy, helped to prepare a well informed management plan, lobbied, made alliances and pressed for implementation of basic conservation measures, and did the monitoring for the evaluation of management on its own resources. Emerging ecotourism helped to raise the awareness of the local stakeholders and to create socio-economic benefits from nature conservation. However, from the taxonomic point of view, the focus was on birds of prey, while cultural and financial restrictions did not permit an extensive implementation of measures for other taxa.

Also the Bulgarian share of the study area was sparcely populated and maintained almost undisturbed biodiversity. Unfortunately, during the period between the end of the old regime and the implementation of the pan-european environmental conservation measures, a significant part of the wild habitats and species suffered from tremendous reduction. For instance, between 1992 and 2000, 70% of the riverain forests along the Maritza River (the biggest river in southern Bulgaria) was cut down and consequently, the local populations of colonially breeding birds decreased by 70% (Green Balkans, pers. obs.). Although a “National Strategy on the Biodiversity” was developed in 1993–1994, all European Conventions in the field of environmental protection were ratified in 1990–1997, and nine national laws related to nature conservation were announced in 1997–1999, environmental protection remains ineffective. The main reasons for biodiversity loss in the area could be attributed to poverty and to land privatisation from 1996. Many nature-protective NGOs (Bulgarian Society for the Protection of Birds, Green Balkans, WWF, Bulgarian Biodiversity Foundation, etc.) fight against these negative environmental processes, but still major problems remain including the lack of developed capacity of environmental policy makers, especially at the administrative level. Only 5% of the personnel in the Bulgarian Ministry of Environment and Waters (MEW) works in the field of nature conservation and only one to three biodiversity experts work in each of the 15 regional inspections of the MEW, therefore one expert is responsible for the environmental monitoring, threat control and for the effectiveness of seven international conventions and 19 national legislative acts of an area of 5000 km^2.

Implementation rates of different categories of recommendations

Regarding the categories of implementation, Greece has generally higher implementation values than Bulgaria. Comparatively high values in both countries were achieved for tourism and environmental education, while very low values were achieved in both countries for recommendations regarding legislation. Greece obtained the highest rates for the categories hunting and fishing, and administration and regulation. Although the Greek rates are still considered as unsatisfactory, they may be attributed to the relative effectiveness of the Greek Forest Service in certain issues, a quality which however declined recently as a result of political decisions and financial restrictions. The categories where Bulgaria obtained comparatively good rates were landscape conservation and tourism and environmental education. The reason could be socio-economic as landscape conservation is directly related to ecotourism development, an issue considered as an important factor for the progress of the local economy (Gerasimov & Stoeva, 1997). Landscape conservation also might be a closer concept to the public than the protection of specific taxa.

Conclusions and recommendations

Science has long identified many problems. What is missing is the political will to improve environmental conservation and create a decent system of protected areas in the countries, capable of coping with site-specific issues. NGOs and independent researchers who have been working in the area for long are able to suggest a number of interconnected priority issues at national, regional and local level. These multi-scale issues must be tackled in a parallel way to create the necessary framework for a satisfactory conservation of biodiversity hot spots in the Eastern Rhodopes (Catsadorakis et al., 2010).

We have to conclude that both in Greece and Bulgaria, scientists should shift their research focus from purely descriptive to applied ecological and conservation science (Pullin et al., 2009; Poirazidis et al., 2010c). They should further be encouraged to communicate their main research findings to the responsible authorities through native language texts. Authorities should provide incentives for increasing the access to scientific literature e.g. by promoting open access journals or by covering the costs of access to standard journals for conservation authorities, and they should promote participatory approaches and effective communication strategies (EPBRS, 2009, 2010; Papageorgiou & Vogiatzakis, 2006) such as web pages in local languages, which organise and prioritise solutions without complex scientific argumentation.

We can further conclude from our research that for successful nature conservation, there is an urgent need for high standard conservation relevant research for many taxa (including rare and rarely studied ones). Additionally, there is a tremendous need for an increase of quality and quantity of evaluations regarding the effectiveness of conservation measures, as well as for the establishment, funding and staffing of public authorities. Crucial are also cross-border conservation and management activities such as joint projects and conservation initiatives. The European Green Belt is a very important instrument to achieve these prerequisites as it is serving as originator and promoter for several of the recommended activities (Terry et al., 2006), and not least for initiating and enhancing collaborations for

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nature conservation on the scattered political map of southeastern Europe.

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Appendices A–D. Supplementary data

Supplementary data associated with this article can be found, in the online version, at doi:10.1016/j.jnc.2011.01.001.

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Appendix A. Questionnaire filled by local conservation experts assessing the implementation of conservation recommendations in the Eastern Rhodopes Mountains.

Recommendation Nr: XY

1. Have you ever heard before about this recommendation? (yes/no/don't remeber)

1.1 If YES, please choose from where you were informed for the first time (1 option)

a) existing management plans
b) expert opinion from expert(s) outside your team
c) books, handbooks
d) documentation or personal accounts of traditional management practices
e) scientific publications
f) reviews in scientific journals
g) unpublished reports
h) popular articles (incl. media such as TV)
i) internet
j) own research
k) other

2. Is this recommendation implemented in your area? (yes/no/don't know)

2.1 If you awnsered **NO**:

2.1.1 Why not? (mark all options which are "true")

a) There is no such authority responsible to implement this recommendation
b) The responsible authority is not competent enough to implement such recommendation
c) The responbile authority implemented inadequately this recommendation
d) The recommendation is not known in my area
e) The recommendation is not considered as important for my area
f) The authority does not implement it as it is considered to have negative side effect on other aspects of nature/ecosystems/biodiversity
g) The authority does not implement it because is is supposed to have negative efects on human activities in the area (hunting, agriculture, forestry…)
h) The implementation of the recommendation is expensive
i) other:

2.1.2 Will it be implemented in the near future? (yes/no/don't know)

2.2 If you awnsered **YES**:

2.2.1 By whom?

2.2.2 When (year)

2.2.3 Has justification been given for its implementation? (yes/no/don't know)

2.2.3.1 If **yes,** in the form of: (1 option only)

a) existing management plans
b) expert opinion from expert(s) outside your team
c) books, handbooks
d) documentation or personal accounts of traditional management practices
e) scientific publications
f) reviews in scientific journals
g) unpublished reports
h) popular articles (incl. media such as TV)
i) internet
j) own research
k) other:

2.2.4 Is this action a continuation of traditional practices?

2.2.5 Is the effectiveness of the implemented recommendation evaluated in your area? (yes/no/don't know)

2.2.5.1 If NO, why not? (mark all options which are "true")

a) There is no such authority responsible to evaluate this implementation
b) The responsible authority is not competent enough to evaluate such implementation
c) The responbile authority evaluated inadequately this recommendation
d) The evaluation is not considered as important for this implemented recommendation
e) The authority does not evaluate this implemented recommendation, because it is afraid of results that would suggest ineffectivness
f) The evaluation of the effectivness of the implemented recommendation is expensive
g) Potential effects of the implemented recommendation will be detectable only after many years, and it is still too early to evaluated them
h) other:

2.2.5.2 Will it be evaluated in the near future? (yes/no/don't know)

Appendix B. Number of articles,number of conservation recommendations, and implemention rate per taxon, for the Greek (GR) and the Bulgarian (BG) part of the Eastern Rhodopes Mountains.

Taxon	**Number of articles**		**Number of articles containing recommendations**		**Number of recommendations**		**% of implemention**	
	GR	**BG**	**GR**	**BG**	**GR**	**BG**	**GR**	**BG**
Animalia								
Birds	51	10	34	10	173	188	56.8	10.1
Black Vulture	15	-	10	-	59	-	76.3	-
Short-toed Eagle	6	-	5	-	16	-	37.5	-
Lesser-spotted eagle	3	-	1	-	12	-	33.3	-
Imperial Eagle	-	1	-	1	-	9	-	22.2
Lesser Kestrel	-	1	-	1	-	2	-	0.0
Other birds of prey	8	-	6	-	8	-	14.3	-
Vulture diversity	1		1		17		58.8	
Bird of prey diversity	8		7		51		60.8	
Black Stork	2	-	2	-	10	-	40.0	-
Shrikes	1	-	1	-	2	-	0.0	-
Landbirds / Passerines	7	-	1	-	3	-	0.0	-
various spp./ bird diversity	3	8	0	8	0	177	na	9.6
Mammals	9	7	3	6	3	47	0.0	29.8
bats	5	3	3	2	3	19	0.0	26.3
wolfs	-	1	-	1	-	12	-	16.7
large mammals	-	1	-	1	-	8	-	62.5
small mammals	2	2	0	2	0	8	na	25.0
other	1	-	0	-	0	-	na	-
Amphibians	9	5	2	3	5	11	20.0	27.3
Reptiles	12	11	4	6	8	19	12.5	21.1
Fish	0	2	-	2	-	21	-	9.5
Invertebrates	13	44	3	7	12	35	41.6	17.3
Orthoptera	9	-	2	-	7	-	57.1	-
Butterflies & moths	3	4	1	2	5	18	20.0	11.1
Dragonflies	-	2	-	1	-	3	-	33.3
Other insects	1	23	0	1	0	8	na	25.0
Spiders	-	3	-	1	-	2	-	50.0
Other invertebrates	1	15	0	2	4		na	25.0
Plantae	27	5	6	1	14	26	6.7	15.4
Orchids	7	-	2	-	10	-	10.0	-
other flowering plants	3	-	1	-	1	-	0.0	-
Trees and Shrubs / Woody vegetation	19	-	4	-	4	-	0.0	-
Plant diversity	4	5	0	1	0	26	0.0	15.4

Appendix C. List of articles of the Greek part of the Eastern Rhodopes Mountains that contained conservation recommendations. Year of publication and number of extracted recommendations (R). **[only for electronical supplementary material]**

Year	R	Reference
GREECE		
1971	22	Hoffmann, L., Bauer, W., & Müller, G. (1971). *Proposals for Nature Conservation in Northern Greece*. IUCN.
1979	30	Hallmann, B. (1979). *Guidelines for the conservation of Birds of Prey in Evros*. Morges: IUCN & WWF.
1984	5	Χανδρινός, Γ., & Hallmann, B. (1984). *Οικοανάπτυξη στο Νομό Έβρου: Δέλτα Έβρου – Δάσος Δαδιάς (Σουφλίου)*. Athens: Υφυπουργείο Νέας Γενιάς και Αθλητισμού.
1985	2	Helmer, W., & Scholte, P. (1985). *Herpetological research in Evros, Greece. Proposal for a biogenetic reserve*. Arnhem/Nijmegen: Societas Europaea Herpetologica.
1989	9	Blachos, C. (1989). *Ecology of Lesser Spotted Eagle in Dadias Forest of Evros Prefecture*. Ph.D. thesis, Aristotle University of Thessaloniki.
1989	7	Dennis, R. (1989). *The Conservation and Management of Birds of Prey and their Habitats in Evros; Greece*. Munlochy: The Royal Society for the Protection of Birds.
1991	5	Spiropoulou, S. (1991). *Black Vulture Conservation and Forest Management in Evros, Greece*. MSc. dissertation, University College London.
1995	2	Kalopissis, J. (1995). Cephalanthera epipactoides Fisch & C. A. Meyer. In: D. Phitos, A. Strid, S. Snogerup & W. Greuter (Eds.), *The Red Data Book of Rare and Threatened plants of Greece* pp. 176-177. Athens: WWF Greece.
1995	18	Adamakopoulos, T., Gatzoyannis, S., & Poirazidis, K. (1995). Specific environmental study of the Forest of Dadia. Part C. Athens: Ministry for the Environment, Physical Planning and Public works , Ministry of Agriculture & WWF Greece.
1995	1	Christensen, K. I. (1995). *Eriolobus trilobatus* (Poiret) M. Roemer Rosaceae. In: D. Phitos, A. Strid, S. Snogerup & W. Greuter (Eds.), *The Red Data Book of Rare and Threatened plants of Greece* pp. 254-255. Athens: WWF Greece.
1995	1	Kamari, G. (1995). *Minuartia greuteriana* Kamari, Caryophyllaceae. In: D. Phitos, A. Strid, S. Snogerup & W. Greuter (Eds.), *The Red Data Book of Rare and Threatened plants of Greece* pp. 362-363. Athens: WWF Greece.
1996	3	Poirazidis, K., Skartsi, Th., Pistolas, K., & Babakas, P. (1996). Nesting habitat of raptors in Dadia reserve, NE Greece. In: J. Muntaner & J. Mayol (Eds.), *Biology and Conservation of Mediterranean Raptors,1994.* pp. 327-333. SEO/Birdlife.
1996	17	Skartsi, Th. (1996). An integrated management project in Dadia Forest Reserve implemented by WWF Greece. In: P. Regato (Ed.), *SILVA Handbook 1996. A WWF course about Mediterranean forests* pp. 107-114. Rome: WWF Mediterranean.
1996	9	Vlachos, C., & Papageorgiou, N. (1996). Breeding biology and feeding of the lesser-spotted eagle (Aquila pomarina) in Dadia Forest, north-eastern Greece. In: B.-U. Meyburg & R. D. Chancellor (Eds.), *Eagle Studies* pp. 337-347. Berlin: World Working Group on Birds of Prey and Owls.
1998	2	Bakaloudis, D. E., Vlachos, C.G., & Holloway, G. J. (1998). Habitat use by short-toed eagles (*Circaetus gallicus*) and their reptilian prey during the breeding season in Dadia Forest (north-eastern Greece). *Journal of Applied Ecology, 35,* 821-828.
1998	3	Capper, S. (1998). *The predation of Testudo spp. by Golden Eagles Aquila chrysaetos in the Dadia forest reserve, NE Greece*. MSc. dissertation, Reading University.
1998	6	Hallmann, B. (1998). The Black Vultures of Greece. In: E. Tewes, J. J. Sanchez, B. Heredia & M. Bijleveld van Lexmond (Eds.), *International Symposium on the Black Vulture in South Eastern Europe and Adjacent regions* pp. 27-32. Palma de Mallorca: Black Vulture Conservation Foundation & Frankfurt Zoological Society.
1998	17	Katsadorakis, G., Poirazidis, K., Gatzogiannis, S., Adamakopoulos, T., Tsekouras, G., & Matsouka, P. (1998). The management of the vulture population and habitat in Dadia Forest Reserve (Greece): a conceptual framework. In: E. Tewes, J. J. Sanchez, B. Heredia & M. Bijleveld van Lexmond (Eds.), *International Symposium on the Black Vulture in South Eastern Europe and Adjacent regions* pp. 11-18. Palma de Mallorca: Black Vulture Conservation Foundation & Frankfurt Zoological Society.
1998	5	Spyropoulou, S. (1998). Black Vulture Conservation in the Dadia forest Reserve. Actions taken up to 1992. In: E. Tewes, J. J. Sanchez, B. Heredia & M. Bijleveld van Lexmond (Eds.), *International Symposium on the Black Vulture in South Eastern Europe and Adjacent regions* pp. 33-38. Palma de Mallorca: Black Vulture Conservation Foundation & Frankfurt Zoological Society.

1998 4 Valaoras, G. (1998). Alternative development and biodiversity conservation: Two case studies from Greece. *The Mountain Forum Online Library*, http://www.mtnforum.org/rs/ol.cfm.

1998 1 Vlachos, C., Papageorgiou, N., & Bakaloudis, D. (1998). Effects of the feeding station establishment on the Egyptian vulture *Neophron percnopterus* in Dadia forest, North Eastern Greece. In: R. D.Chancellor, B.-U. Meyburg & J. J. Ferrero (Eds.), *Holartic Birds of Prey* pp. 197-207. Badajoz: World Working Group on Birds of Prey and Owls & ADENEX.

2000 3 Bakaloudis, D. E. (2000). *The ecology of Short-toed Eagle (Circaetus gallicus, Gm.) in Dadia-Lefkimi-Soufli Forest complex, Thrace, Greece*. Ph.D. thesis, Reading University.

2000 1 Ivanova, Th. (2000). New data on bats (Mammalia: Chiroptera) from the Eastern Rhodopes, Greece (Thrace, Evros). *Historia naturalis bulgarica*, *11*, 117-125.

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Appendix D. List of conservation recommendations extracted in this study, conservation goal and reference area (DNP = Dadia- Lefkimi-Soufli Forest National Park; ER = Eastern Rhodopes; PA = Protected Area). **[only for electronical supplementary material]**

Recommendation	Goal	Area
GREECE		
Legislation		
Establishment of legal/administrative regulations	Integrated conservation management	Buffer zone
Only temporal forest tracks should be allowed to be built in the buffer area	Maintain valuable natural resources	Buffer zone
Forbiddance of creation of new roads in the core area	Integrated conservation management	Core area
Forbiddance of hunting and creation of new roads	Black vulture conservation	Core area
Forbiddance of any activity in the core area from 1st March until 15 September	Black vulture conservation	Core area
Strict control of access to the core area to avoid disturbance to Black vulture and other species	Black vulture conservation	Core area
Greece should sign the international *Agreement on the Conservation on Bats in Europe* (EUROBATS)	Bat conservation	Everywhere
Seasonal restrictions for logging, hunting and other human activities	Integrated conservation management	Whole DNP
Implementation of relevant laws to stop poisoning events (the use of poisoned baits)	Reduce poisoning events on vultures	Whole DNP
Secure protection, implement conservation measures	Integrated conservation management	Whole DNP
Progress in legal protection	Improvement of the functioning of the reserve	Whole DNP
Avoidance of disturbance during breeding period of Lesser spotted eagle: forbiddance of military activities	Lesser spotted eagle conservation	Whole DNP
Protection of nest sites	Black vulture conservation	Whole DNP
Administration and Regulation		
Research centre defines and/or proposes new regulations	Research for conservation	"Evros Park"
Submit projects to intergovernmental organizations	International collaboration	"Evros Park"
Have well-informed experts as custodians of the park	Administration and wardening	"Evros Park"
Establishment of a research station in the Park	Research for conservation	"Evros Park"
Stimulate interchange of knowledge and work between Greek and foreign experts in research, management, legislation, education and public relations	International collaboration	"Evros Park"
Establishment of a research station in the Park	Research for conservation	"Evros Park"
Second buffer zone towards the east until Evros	Maintain valuable natural resources	"Wider area"
Enforce and broaden the transborder collaboration in bat conservation	Large-scale, integrated bat conservation	"Wider area"
Establish a communication network	Integrated conservation management	"Wider area" 1500km2
Road access in significant parts of the buffer zone must be controlled using bars managed by the Forest Service.	Black vulture conservation	Buffer zone
Delineation of the buffer zone	Integrated conservation management	Buffer zone
Establishment of a National Park	Maintain valuable natural resources	Buffer zone
Completed management plan	Integrated conservation management	Buffer zone
Conservation of a “sensibility zone” area	Raptor conservation	Buffer zone
Establishment of a central Reserve Authority	Integrated conservation management	Buffer zone
Research staff should be member of conservation committee	Maintain valuable natural resources	Core area
Controlling and guarding of the core area by Forest Service	Integrated conservation management	Core area

Preservation and control of road system in the core area	Black vulture conservation	Core area
Creation and implementation of general regulations and restrictions	Integrated conservation management	Core area
Maintenance and control of the road network	Integrated conservation management	Core area
Determination of high sensibility area	Black vulture conservation	Core area
Wardening the core zones	Vulture conservation	Core area
Practical and legal support for wardens	Maintain valuable natural resources	Core area
Determination of high sensitivity area	Integrated conservation management	Core area
Organization of fire-prevention plan	Integrated conservation management	Core area
Protection of nest sites in the core area	Black vulture conservation	Core area
Intensify protection, training of wardens	Increase protection of vulture nests	Core area
Superintendence and staff should manage the reserve	Maintain valuable natural resources	Core area
Protected area managers have to avoid value contradictions while defining or communicating management priorities as well as during the preparation of environmental awareness campaigns and environmental education projects.	Integrated conservation management	Everywhere
Fundraising	Integrated conservation management	Everywhere
Lobbying for further protection	integrated conservation management	Everywhere
Enforce and broaden the transborder collaboration in vulture conservation	Vulture conservation	Evros
Enforce and broaden the transborder collaboration in vulture conservation	Black vulture conservation	Greece, Bulgaria
Enforce and broaden the transborder collaboration in vulture conservation	Black vulture conservation	Greece, Bulgaria, Balkans
Presence of a permanent scientific authority in the Park	Enhance butterfly diversity	Whole DNP
Promotion of a systematic anti-poisoning strategy, at local and national level	Reduce poisoning events on vultures	Whole DNP
Collaboration of WWF and management body	Black vulture conservation	Whole DNP
Establishment of a central Reserve Authority	Integrated conservation management	Whole DNP
Cluster model of 2 core areas and 1 buffer zone	Maintain valuable natural resources	Whole DNP
Establishment of local conservation agency	Improvement of the functioning of the reserve	Whole DNP
Establishment of a protected area (National Park)	Maintain valuable natural resources	Whole DNP
Park provided with staff for management and research	Maintain valuable natural resources	Whole DNP
Establishment of a central Reserve Authority	Improvement of the functioning of the reserve	Whole DNP
Research centre defines and/or proposes new regulations	Maintain valuable natural resources	Whole DNP
Establish a committee for conservation issues for the whole prefecture	Maintain valuable natural resources	Whole DNP
Collaboration between local community, state authorities and NGO (WWF Greece)	Ecotourism and NP management	Whole DNP
Creation of integrated management plan	Integrated conservation management	Whole DNP
Establishment of a protected area (National Park)	Maintain valuable natural resources	Whole DNP
Creation of a biogenetic reserve for birds of prey	Raptor conservation	Whole DNP
Research		
Analyze landscape change, land management and its effects	Landscape monitoring & evaluation of measures	Buffer zone
Research on thresholds of forest openings size that do not harm forest species and regeneration	Sustainable forest management	Core area
Assess demographics and gene flow of 2 Spotted eagle species	Spotted eagle conservation	E Europe

Use of complementarity approach in reserve design	High quality reserve design	Everywhere
Use of complementarity approach in reserve design	High quality reserve design	Everywhere
Use of different approaches in reserve design	High quality reserve design	Everywhere
Apply the theoretical methodology developed by Papageorgiou et al. (2006)	Conserve forest genetic diversity	Everywhere
Experimental introduction of European Suslik in selected areas	Long-legged Buzzard conservation	Evros
Habitat suitability modelling for further indicator organisms	Sustainable forest management	Managed forests
Inclusion of further taxa into habitat suitability modelling	Sustainable forest management	Managed forests
Pesticide impact on raptors	Raptor conservation	Whole DNP
Habitat suitability modelling for selected groups of organisms to develop management scenarios for managed forests	Sustainable forest management	Whole DNP
Future study for long term movement pattern of the tortoises' females	Reptile conservation	Whole DNP
Further research on butterfly species of European Conservation Concern	Conservation of endangered species	Whole DNP
To investigate where Egyptian vultures find their tortoise prey in the National Park	Egyptian vulture conservation	Whole DNP
Study possible pesticide impact on Short-toed eagle	Short-toed eagle conservation	Whole DNP
Studies of food requirements and availability for Black vulture should be initiated	Vulture conservation	Whole DNP
Further examine new requirements and the potentials for ecotourism that are derived by the designation of the Protected Area as a NP	Knowledge on potential of ecotourism	Whole DNP
Research on availability of suitable Golden eagle nest site	Raptor conservation	Whole DNP
Assess the role of fire on the region's ecosystems	Orchid conservation	Whole DNP
Further research on causes and circumstances of butterfly decline in the DNP	Butterfly conservation	Whole DNP
Further research on biodiversity indicators	Sustainable forest management	Whole DNP
Specification and mapping of favourably area for breeding of Lesser spotted eagle	Raptor conservation	Whole DNP
Assess availability of suitable raptors’ nesting sites	Raptor conservation	Whole DNP
Assess important areas for breeding raptors	Raptor conservation	Whole DNP
Nest site preference of Black stork outside the core area	Black stork conservation	Whole DNP
Creation of a specific environmental study	High quality environmental management	Whole DNP
Recognition of environmental variables which are necessary for Black vulture's reproduction in the area	Vulture conservation	Whole DNP
Investigation of genetic diversity of Black vulture population	Vulture conservation	Whole DNP
Configuration of methodology for the systematic monitoring of landscape and biotopes	Systematic monitoring of landscape and biotopes	Whole DNP
Configuration of methodology for the systematic monitoring of Black vulture	Systematic monitoring of Black vulture	Whole DNP
Regular checking of endangered plant's conservation status	Conservation of endangered species	Whole DNP
Configuration of methodology for the systematic monitoring of raptors	Systematic monitoring of raptors	Whole DNP
Monitoring		
Long-term landscape monitoring	Integrated conservation management	"Wider area" 1500km2
Monitoring in buffer zone	Integrated conservation management	Buffer zone
Suitable Black vulture habitat monitoring following systematic monitoring plan	Vulture conservation	Buffer zone
Creation of a monitoring plan for the core area	Integrated conservation management	Core area

Scientific evaluation of wildlife and the evolution of forest vegetation after the implementation of the proposal measurements	Evidence based conservation management	Core area
The gradual canopy closure of the forest around Black vulture nest sites must be monitored periodically	Vulture conservation	Core area
Long-term landscape monitoring	Landscape conservation	Whole DNP
Monitoring of poisoning events	Vulture conservation	Whole DNP
Establishment of a monitoring scheme on biological and environmental factors	Integrated conservation management	Whole DNP
Systematic monitoring of raptor populations	Raptor conservation	Whole DNP
Monitoring plan of Black vulture	Vulture conservation	Whole DNP
Marking of Black vulture	Vulture conservation	Whole DNP
More effort in monitoring Black vulture nest sites and productivity	Vulture conservation	Whole DNP
Long-term monitoring programme for the Short-toed eagle population	Short-toed eagle conservation	Whole DNP
Telemetry of Black vulture	Vulture conservation	Whole DNP
Landbird community included as indicator species in monitoring programme	High quality biodiversity monitoring	Whole DNP
Systematic monitoring of raptor populations	Raptor conservation	Whole DNP
Evaluation of management effects on forest and wildlife	Evidence based conservation management	Whole DNP
Use of woody plants as indicators in biodiversity monitoring	High quality biodiversity monitoring	Whole DNP
Herpetofauna should be integrated per se in the management and monitoring scheme of NP	Herpetofauna conservation	Whole DNP
Long-term landscape monitoring	Landscape conservation	Whole DNP
the use of 9 Orthoptera indicator species and *Paranocarodes chopardi* in the reserve monitoring program	High quality biodiversity monitoring	Whole DNP
Landscape conservation		
Preservation of landscape heterogeneity	Black vulture conservation	Buffer zone
Maintenance of forest openings in the buffer zone	Orthoptera conservation	Buffer zone
Maintain rural mosaics and forest openings in the buffer zone	Bird conservation	Buffer zone
Improvement of the water conditions of the forest	Integrated conservation management	Core area
Improvement of hydrous condition for forest (for example, ponds)	Integrated conservation management	Core area
Maintenance of forest openings in the core area	Orthoptera conservation	Core area
Management of rural areas	Integrated conservation management	Core area
Support (reintroduce) wild ungulates	Maintenance of forest openings for Short-toed eagle conservation	Grasslands in DNP
Maintenance of rural mosaic landscape	Bird conservation	Mediterranean
Maintain human activities that support landscape mosaic	Short-toed eagle conservation	Whole DNP
Conserve the present gradient from open vegetation to more closed woodland	Raptor conservation	Whole DNP
Maintenance of forest heterogeneity	Orthoptera conservation	Whole DNP
Maintenance of forest openings with high vegetation diversity and structural complexity	Maximizing biodiversity conservation	Whole DNP
Maintenance of forest heterogeneity	Orthoptera conservation	Whole DNP
Maintain rural mosaics and forest openings in the whole park	Maximizing biodiversity conservation	Whole DNP
Controlled burning in the winter time	Conservation of reptiles and raptors	Whole DNP
Maintenance of forest openings with high vegetation diversity and structural complexity	Maintenance of vegetation diversity	Whole DNP
Support (reintroduce) wild ungulates	Maintenance of forest openings for orchid conservation	Whole DNP

Creation and/or restoration of small forest openings in dense forest areas, through low-intensity livestock grazing and re-introduction of herd grazing	Maintenance of forest openings for raptor conservation)	Whole DNP
Maintenance of habitat heterogeneity	Conservation of biodiversity	Whole DNP
Agriculture and livestock rearing		
Control new agricultural development	Sustainable agriculture	"Evros Park"
Support of traditional and alternative forms of agriculture and livestock raining	Integrated conservation management	"Wider area" 1500km2
Enhancement of stock farming	Integrated conservation management	Core area
Maintain traditional grazing system in the core area	Black vulture conservation	Core area
Encourage grazing in the core area	Integrated conservation management	Core area
Maintain traditional grazing system in the core area	Integrated conservation management	Core area
Permission for livestock grazing, agriculture, research in the core area	Integrated conservation management	Core area
Management of agricultural areas in the core area	Sustainable agriculture	Core area
Prevent rural depopulation and abandonment of traditional stock-raising practices in Evros	vulture conservation	Evros
Improvement of traditional agriculture practice at small scale	Herpetofauna conservation	Evros
Low-intensity farming systems supported through an active EU agricultural policy.	Short-toed eagle conservation	Farmlands in DNP
Minimize the use of agrochemicals in the intensively cultivated land	Short-toed eagle conservation	Farmlands in DNP
Grazing concentrated on dense shrublands and, to a lesser extent, open areas.	Short-toed Eagle conservation	Grasslands in DNP
Encourage grazing in the whole NP	Raptor conservation	Whole DNP
Encourage grazing in the whole NP	Short-toed eagle conservation	Whole DNP
Adopt a pluri-annual rotational grazing system	Maintain open areas for orchid conservation	Whole DNP
Favour the use of goats and sheep instead of cows	Maintain open areas for orchid conservation	Whole DNP
Encouragement of traditional agricultural practices in areas surrounded by forest and low-intensity grazing	Butterfly conservation	Whole DNP
Encourage grazing in the whole NP	Lesser spotted eagle conservation	Whole DNP
Enhancement of periodical livestock grazing	Orthoptera conservation	Whole DNP
Enhancement of periodical livestock grazing	Preservation of landscape heterogeneity for Orthoptera conservation	Whole DNP
Forest management		
Forestry practices that maintain value of woodland for scenic beauty and wildlife	Sustainable forest management	"Evros Park"
Systematic rehabilitation of more degraded woodlands	Sustainable forest management	"Evros Park"
Forestry based on indigenous species	Sustainable forest management	"Evros Park"
No logging of trees in neighborhood of raptor/Black stork nests	Raptor and Black stork conservation	"Evros Park"
Felling projects and road tracing in buffer zone should be studied and approved by supervisor	Maintain valuable natural resources	Buffer zone
Logging activities and other disturbance in buffer zone must be restricted to the autumn period	Black vulture conservation	Buffer zone
Long term restoration of afforested areas to mixed self-sustaining forests including high oak forest	Raptor conservation	Buffer zone
Measures and limitations for forest exploitation	Integrated conservation management	Buffer zone
Felling in buffer zone should stay away 50 m from nest sites	Conservation of endangered birds	Buffer zone
Regulations for forest and logging exploitation in the buffer zone	Integrated conservation management	Buffer zone

Preservation of suitable nest trees for Black vulture in the buffer area	Black vulture Conservation	Buffer zone
Reforestation prohibited in the buffer zone	Conservation of endangered birds	Buffer zone
Creation of an annual logging catalogue for the buffer area	Sustainable forest management	Buffer zone
Preservation of all remaining old tree stands in the core area	Preservation of natural forests	Core area
Forest exploitation forbidden in the core area	Black vulture conservation	Core area
Seasonal restrictions for logging	Sustainable forest management	Core area
Management of stands and of partially wooded areas in the core area	Sustainable forest management	Core area
Use "soft-forestry" methods; pay workers by the hour and not per wood quantity	Preservation of natural forests	Core area
Setting aside the previously untouched woodland in the core area	Conservation of endangered birds	Core area
Organization of the collection, transportation and distribution of timber products in the core area	Sustainable forest management	Core area
Stop plantation forestry in the core area	Preservation of natural forests	Core area
Forest exploitation forbidden in the core area	Conservation of endangered birds	Core area
Spatial and temporal organization of interventions	Sustainable forest management	Core area
Preservation of all remaining old tree stands in the core area	Black vulture Conservation	Core area
Maintenance of ecologically acceptable programme of selective felling/natural regeneration in managed woods	Raptor conservation	Core area
Measurements of vegetation management (logging plan)	Integrated conservation management	Core area
Prohibition of mature tree cutting in the core area	Black vulture conservation	Core area
Organization of a fire-prevention protection plan	Integrated conservation management	Core area
Preservation of the old oak forest in the low and high mountain	Herpetofauna conservation	Evros
Mitigation of afforestation in marginal fields	Short-toed eagle conservation	Farmlands in DNP
Spatial forest management planning (aimed at sustainability)	Sustainable forest management	Managed forests
Preservation of all remaining old tree stands in the whole park	Raptor conservation	Whole DNP
Preservation of small groups of mature forest with loose density and sparse intermediate canopy	Black stork conservation	Whole DNP
Protection of mature pine trees around 60-70 years old	Lesser spotted eagle conservation	Whole DNP
Supply foresters with information & directions that will assist them in the proper planning of forest works	Raptor conservation	Whole DNP
Increase of forest vegetation with implementation of suitable forest measurements (regulation of grazing and reforestation)	Sustainable forest management	Whole DNP
Preservation of all remaining old tree stands in the whole park	Black stork conservation	Whole DNP
Preservation of isolated groups of mature trees in each forest stand	Black stork conservation	Whole DNP
Preservation of isolated groups of mature trees in each forest stand	Black stork conservation	Whole DNP
Maintenance of the open forest structure	Herpetofauna conservation	Whole DNP
Forestry operations should be restricted during the breeding season in an area of 800m radius around existing nests	Black vulture conservation	Whole DNP
Preservation of mature pines close to water	Black stork conservation	Whole DNP
Logging activities applied in the end of the Black stork breeding season and/or after the young fledged from nests	Black stork conservation	Whole DNP
Preservation of tall trees on steep mountain slopes	Black vulture conservation	Whole DNP
Preservation of all remaining old tree stands in the whole park	Goshawk conservation	Whole DNP
Strictly avoid any forestry activity in the vicinity of Black vulture nests and during breeding season	Black vulture conservation	Whole DNP

Logging of small tree groups in homogenous forest	Black stork conservation	Whole DNP
Maintain or reduce rate and duration of timber extraction in the Park	Short-toed eagle conservation	Whole DNP
Collaboration between foresters and conservationists	Black vulture conservation	Whole DNP
Preservation of all remaining old tree stands in the whole park	Raptor conservation	Whole DNP
Wildlife management		
Cultivation of *Eriolobus trilobatu*s in botanical gardens	Tree species conservation	Botanical gardens
Urgent actions are needed in the buffer zone	Integrated conservation management	Buffer zone
Creation of an action plan for Black vulture conservation	Vulture conservation	Buffer zone
Protection of the active vulture nesting sites	Vulture conservation	Core area
Prevent demolition of disused mine's entrances	Bat conservation	Core area
Operation of the feeding system for vultures	Vulture conservation	Core area
Application of the management plan in the core area	Effective management plan	Core area
Reintroduction / reinforcement plan for ungulates (deer) and of mid-sized birds (partridges) in the core area	Fauna conservation	Core area
Reintroduction and demographic enhancement of *Lepus europaeus, Perdix perdix, Alectoris chukar, Dama dama, Capreolus capreolus*	Fauna conservation	Core area
Creation of artificial feeding places for vultures	Vulture conservation	Core area
Avoid Black vulture restocking	Vulture conservation	Europe
Consider genetic issues in the implementation of European conservation strategies for Black vulture	Vulture conservation	Europe
Promoting hedges containing shrubs and trees in case of land reallotment	Maintain/improve habitat for Short-toed eagle	Farmlands in DNP
Water spring cultivation or the construction of small ponds near openings	Improve habitat for Short-toed eagle & herpetofauna	Grasslands in DNP
Water management to favour water concentrations in grasslands	Improve habitat for Short-toed eagle & herpetofauna	Grasslands in DNP
Total protection of the natural habitat of *Cephalanthera epipactoides* (rare orchid)	Orchid conservation	Greece
Fencing of one or two areas of high abundance of *Cephalanthera epipactoides* (rare orchid) away from tourists	Orchid conservation	Greece
Assure that grazing only takes place after the orchids flowering season (after may)	Orchid conservation	Whole DNP
Prevent construction of forest roads near existing Black vulture nest sites	Vulture conservation	Whole DNP
Creation of small wetlands in the forested area	Black stork conservation	Whole DNP
Preservation of scrubland in pine forest for protection of small fauna (reptiles, rodents)	Lesser Spotted eagle conservation	Whole DNP
Network of artificial feeding places for Black vultures	Vulture conservation	Whole DNP
Increasing of natural food sources for Black vultures	Vulture conservation	Whole DNP
Organization of a fire-prevention protection plan	Black vulture conservation	Whole DNP
Conservation management for birds of prey in Dadia	Raptor conservation	Whole DNP
Dispersion of suitable Black stork nesting habitat	Black stork conservation	Whole DNP
Preservation of old olive, almond and walnut groves, and old *Platanus* along the streams	Masked-shrike conservation	Whole DNP
Sustain a suitable number of nesting sites for Lesser spotted eagle	Lesser spotted eagle conservation	Whole DNP
Detection of all trees with Lesser spotted eagle nests and forbiddance of any human activities around 100m of nest	Lesser spotted eagle conservation	Whole DNP
Creation of new nest sites within the current limits of the Black vulture colony	Vulture conservation	Whole DNP
Management of Short-toed eagle nesting area	Short-toed eagle conservation	Whole DNP
Management of Short-toed eagle foraging areas	Short-toed eagle conservation	Whole DNP
Operation of 2 feeding sites for vultures	Vulture conservation	Whole DNP
Firebreaks as a conservation tool for tortoises' nest site and	Fauna conservation	Whole DNP

eagles' hunting areas		
Delimitation and protection of European ground squirrel areas	Lesser spotted eagle conservation	Whole DNP
Introduction of *Alectoris chukar* (Chukar partridge)	Lesser spotted eagle conservation	Whole DNP
Management for a wide range of species	Effective management plan	Whole DNP
Maintain sheep and goat grazing in the Park	Woodchat shrike conservation	Whole DNP
Habitat management to maintain a high density of reptiles	Reptiles + Lesser-spotted eagle conservation	Whole DNP
Make Black vulture nests more sturdy for storms + construction of artificial eyres in the corners of former distribution area	Vulture conservation	Whole DNP
Operation of 3 feeding sites for vultures	Vulture conservation	Whole DNP
Hunting & fishing		
Sport hunting and fishing in carefully chosen and well-marked zones, away from tourist itineraries	High quality zoning of NP	"Evros Park"
Complete ban on hunting of birds of prey and larger carnivores (exception for problematic animals)	Raptor and Large carnivores conservation	"Evros Park"
Prohibition of hunting in the core area	Avoid negative human impact in core area	Core area
Forbiddance of hunting in the core area	Integrated conservation management	Core area
Stop the use of poisoned baits	Vulture conservation	Everywhere
Stop the use of poisoned baits	Black vulture conservation	Everywhere
Stop the use of poisoned baits	Vulture conservation	Evros
Prohibition of hunting in important raptor areas	Raptor conservation	Whole DNP
Stop the use of poisoned baits	Raptor conservation	Whole DNP
Tourism & environmental education		
Road building should maintain scenic beauty and wildlife, and create accesses to tourist	High quality regional planning	"Evros Park"
Tourist facilities in locations that do not damage scenic beauty or wildlife	High quality regional planning	"Evros Park"
Research Centre with own research programme and coordinate activities of visitors	High quality NP information and research center	"Evros Park"
Build access and facilities for tourists guided tours	High quality ecotourism	"Evros Park"
Creation of screens and hides for wildlife viewing	High quality ecotourism	"Evros Park"
Prepare and distribute informative material (e.g. guide books and pamphlets)	High quality ecotourism & environmental education	"Evros Park"
Development of soft recreational activities	Sustainable rural development	"Wider area" 1500km2
Regulation in infrastructure and recreation areas	Decrease human impact and human disturbance	Buffer zone
General public should not leave roads, stay overnight, collect organisms, damage nature	Avoid negative human impact	Core area
Organization and control of tours and sight-seeing schemes in the core area	Controlled tourism in core area	Core area
Organization and control of normal and conducted tours	High quality ecotourism & avoid negative impact of unsustainable and uncontrolled tourism	Core area
Limit tourist numbers in the core area	Orchid conservation	Core area
Development of tourism should not be encouraged as long as there is no well managed National Park	Avoid negative impact of unsustainable and uncontrolled tourism	Core area
Organization and control of tours and sight-seeing schemes	Avoid negative impact of unsustainable and uncontrolled tourism	Core area
Creation of infrastructure for tourism in Dadia	Increment of benefits for locals from ecotourism	Dadia village
Combination of tourism and conservation	Sustainable ecotourism as conservation tool	Everywhere

Raise the environmental awareness of rural residents	Environmental education of locals as conservation tool	Everywhere
Thoughtful conservation measures in ecotourism	Sustainable ecotourism	Everywhere
Attraction of visitors outside the area has to provide additional income which otherwise would not be generated	Increment of benefits for locals from ecotourism	Everywhere
Ecotourism should focus on the interplay between society and nature	Public awareness & environmental education	Everywhere
High quality management of ecotourism	Involvement of locals in ecobusiness & sustainable tourism	Everywhere
Basic infrastructure to give incentives to private entrepreneurs	Increment of benefits for locals from ecotourism	Everywhere
Involvement in ecotourism and participation in environmental education programs could not suffice to enhance environmental conservation or quality of life issues within rural communities living in protected areas	Enhance environmental conservation or quality of life issues within rural communities living in protected areas	Everywhere
Visitor information and environmental education not confined to mere descriptions of biodiversity and conservation measures applied within protected areas	High quality ecotourism & environmental education	Everywhere
Establishment of an ecotourism infrastructure, data collection and management system, training of local people	High quality ecotourism & environmental education	Everywhere
High quality of the visitor experience	High quality ecotourism	Everywhere
Environmental education	Increased public awareness as conservation tool	Everywhere
Limit tourist numbers in the whole NP	Butterfly conservation	Whole DNP
Accounting for visitors prior knowledge in developing education methods	High quality ecotourism	Whole DNP
Develop public awareness activities related to Black vultures	Black vulture conservation	Whole DNP
Environmental education of local community	Increased public awareness as conservation tool	Whole DNP
Involvement of local communities	Public awareness & environmental education	Whole DNP
Training of eco-guides	High quality ecotourism	Whole DNP
Presentations and promotional material at national and international meetings	Public awareness & marketing	Whole DNP
Visitor group size compatible with natural and social carrying capacity levels	High quality ecotourism	Whole DNP
Suitable ecotourism measurements	Sustainable ecotourism	Whole DNP
Establish limited areas for tourist visits	Orchid conservation	Whole DNP
Balanced increase of ecotourism	Sustainable ecotourism & increment of benefits for locals from ecotourism	Whole DNP
Involvement and sensitization of stakeholders to stop poisoning events	Vulture conservation	Whole DNP
Create accommodation for visitors	Improved ecotourism	Whole DNP
Prepare and distribute informative material (e.g. guide books and pamphlets)	High quality ecotourism & environmental education	Whole DNP
Education and public awareness	Environmental education as conservation tool	Whole DNP
Avoid mass tourism in the whole NP	Orchid conservation	Whole DNP
Sales of local products and local meals	Involvement of locals in ecobusiness	Whole DNP
Content of on-site interpretation favouring interconnections between natural and human features of protected areas	High quality ecotourism	Whole DNP
Educational programmes structured as to take advantage of the site's attraction to visitors	High quality ecotourism	Whole DNP
Sustainable development		
Building of private houses restricted to villages	Avoidance of human activities with significant negative impact on nature	"Evros Park"
No industrial construction should be permitted inside the NP	Avoidance of human activities with	"Evros Park"

	significant negative impact on nature	
Sustainable use of resources in the "wider area"	Sustainable rural development	"Wider area" 1500km2
Support of crafts and small industries	Support traditional lifestyle	"Wider area" 1500km2
Regulation in the rural exploitation areas	Avoidance of human activities with significant negative impact on nature	Buffer zone
Large scale changes in land use only with proper impact assessment	Controlled and sustainable land use & land use change	Buffer zone
Prohibition of the execution of earthwork altering geomorhological features and natural beauties	Controlled and sustainable land use & land use change	Core area
Environmental control of infrastructure projects in the core area	Conservation of core areas	Core area
Prohibition of excavation of mines and quarries	Controlled and sustainable land use & land use change	Core area
No industrial construction should be permitted inside the core area	Conservation of core areas	Core area
Changes in land use controlled by reserve staff	Controlled and sustainable land use & land use change	Core area
Prohibition of installation of houses and huts in the core area without the permission of the superintendent	Conservation of core areas	Core area
Involvement women in women's cooperative	Sustainable rural development	Everywhere
Evolution of local community enterprises	Sustainable rural development	Everywhere
Successful partnership between the private and public sectors	Sustainable rural development	Everywhere
Collaboration of private and public bodies, women, local community leaders and conservationists	Sustainable and balanced development beneficial for all stakeholders	Everywhere
Promote rural development plans that safeguard vulture conservation and expansion	Black vulture conservation	Evros
Restrictions of land use changes in the Park	Controlled and sustainable land use & land use change	Whole DNP
Promote rural development	Sustainable and balanced development beneficial for all stakeholders	Whole DNP
BULGARIA		
Legislation		
Protection of plane forests along the rivers	Plants and fungi conservation	Whole ER
Prohibition the construction and enlargement of infrastructure and facilities (including wind farms and related electricity transmission networks, sport centres, holyday villages, golf courses, etc.)	Bird conservation	PA Krumovitza
Prohibition the pollution with industrial, toxic, household, construction, and other waste	Bird conservation	PA Krumovitza
Prohibition the extraction of inert materials from the rivers beds	Bird conservation	PA Studen Kladenetz
Prohibition the traffic of vehicles (including organization of motor and automobile competitions, outside the existing road network, forest and agriculture roads)	Bird conservation	PA Studen Kladenetz
Prohibition the plugging, afforestation and change in the hydroelectric facilities (including mini hydroelectronic power-stations), as well as construction of dams, shoots and fishponds along the rivers	Bird conservation	PA Madzharovo
Prohibition of drainage activities and the like in the wetlands and their catchments	Bird conservation	PA Byala Reka
Prohibition the introduction of exotic plant and animal species	Bird conservation	PA Byala Reka
Prohibition the traffic of vehicles (including organization of	Bird conservation	PA Byala

motor and automobile competitions, outside the existing road network, forest and agriculture roads)		Reka
Prohibition the destroy of vegetation along river banks	Bird conservation	PA Byala Reka
Prohibition the change of water streams	Bird conservation	PA Byala Reka
Prohibition all activities that degrade the quality of the habitats or cause disturbance to birds (except on designated places and under conditions in accordance with a management plan)	Bird conservation	PA Byala Reka
Prohibition the extraction of inert materials from the rivers beds	Bird conservation	PA Arda bridge
Prohibition the disposal of toxic waste	Bird conservation	PA Arda bridge
Prohibition the release of unpurified industrial and communal waters	Bird conservation	PA Arda bridge
Roost protection	Bats conservation	Whole ER
Banning the taking away of the pups before they had left the lair	Wolf conservation	Whole ER
Prohibition the extraction of inert materials from the rivers beds	Bird conservation	PA Krumovitza
Prohibition all activities that degrade the quality of the habitats or cause disturbance to birds (except on designated places and under conditions in accordance with a management plan)	Bird conservation	PA Krumovitza
Making failsafe the electricity transmission network	Bird conservation	PA Krumovitza
Prohibition the disposal of organic and other substances leading to eutrophication	Bird conservation	PA Studen Kladenetz
Prohibition the pollution with industrial, toxic, household, construction, and other waste	Bird conservation	PA Studen Kladenetz
Prohibition the construction and enlargement of infrastructure and facilities (including wind farms and related electricity transmission networks, sport centres, holyday villages, golf courses, etc.)	Bird conservation	PA Madzharovo
Prohibition the construction of hydroenergetic facilities	Bird conservation	PA Madzharovo
Prohibition all activities that degrade the quality of the habitats or cause disturbance to birds (except on designated places and under conditions in accordance with a management plan)	Bird conservation	PA Madzharovo
Prohibition the plugging, afforestation and change in the hydroelectric facilities (including mini hydroelectronic power-stations), as well as construction of dams, shoots and fishponds along the rivers	Bird conservation	PA Byala Reka
Prohibition the construction of hydroelectric facilities (including dams, shoots and fishponds along the rivers)	Bird conservation	PA Byala Reka
Prohibition the extraction of inert materials from the rivers beds	Bird conservation	PA Byala Reka
Prohibition the establishment of new and enlargement of existing quarries and mines, destroying rocks	Bird conservation	PA Arda bridge
Prohibition the traffic of vehicles (including organization of motor and automobile competitions, outside the existing road network, forest and agriculture roads)	Bird conservation	PA Arda bridge
The lows in Bulgaria should protect the cave invertebrates in E Rhodopes	Protection of the cave invertebrates.	Whole ER
Application of the provisions of International Conventions and Agreements, related to protection of the environment and biological species	Forest conservation	Whole ER
Prohibition the establishment of new and enlargement of	Bird conservation	PA

existing quarries and mines, destroying rocks		Krumovitza
Prohibition the release of unpurified industrial and communal waters	Bird conservation	PA Krumovitza
Prohibition the traffic of vehicles (including organization of motor and automobile competitions, outside the existing road network, forest and agriculture roads)	Bird conservation	PA Krumovitza
Prohibition the construction and enlargement of infrastructure and facilities (including wind farms and related electricity transmission networks, sport centres, holyday villages, golf courses, etc.)	Bird conservation	PA Studen Kladenetz
Prohibition all activities that degrade the quality of the habitats or cause disturbance to birds (except on designated places and under conditions in accordance with a management plan)	Bird conservation	PA Studen Kladenetz
Prohibition the establishment of new and enlargement of existing quarries and mines, destroying rocks	Bird conservation	PA Byala Reka
Prohibition the construction of water installations, which decrease the water retention features of the forests or the underground waters	Bird conservation	PA Byala Reka
Prohibition the disposal of toxic waste	Bird conservation	PA Byala Reka
Making failsafe the electricity transmission network	Bird conservation	PA Byala Reka
Prohibition the construction and enlargement of infrastructure and facilities (including wind farms and related electricity transmission networks, sport centres, holyday villages, golf courses, etc.)	Bird conservation	PA Arda bridge
Prohibition the construction of hydroelectric facilities (including dams, shoots and fishponds along the rivers)	Bird conservation	PA Arda bridge
Prohibition the disturbance of the natural state of the water areas, water streams, their beds, banks and adjacent territories;	Bird conservation	PA Arda bridge
Aid to the undertaken legislative initiatives for updating the Law of Environmental Protection, the criteria on the same and the regimes for preservation of the protected natural territories.	Forest conservation	Whole ER
Voting Regulations for the application of the Bern Convention	Wolf conservation	Whole ER
Structuring an effective system for the compensation of the farmers affected by the wolf	Wolf conservation	Whole ER
Implementation of conservation legislation	Biodiversity conservation	Whole ER
Prohibition the plugging, afforestation and change in the hydroelectric facilities (including mini hydroelectronic power-stations), as well as construction of dams, shoots and fishponds along the rivers	Bird conservation	PA Krumovitza
Prohibition the disposal of toxic waste	Bird conservation	PA Krumovitza
Prohibition the introduction of exotic plant and animal species	Bird conservation	PA Krumovitza
Prohibition the construction activities on the reservoir banks	Bird conservation	PA Studen Kladenetz
Prohibition the establishment of new and enlargement of existing quarries and mines, destroying rocks	Bird conservation	PA Studen Kladenetz
Prohibition the boat quayage in the nesting areas of vultures, egrets, storks, and cormorants	Bird conservation	PA Studen Kladenetz
Prohibition the construction of dams, shoots and fishponds along the rivers	Bird conservation	PA Madzharovo
Prohibition the disturbance of the natural state of the water areas, water streams, their beds, banks and adjacent territories;	Bird conservation	PA Madzharovo

Prohibition the pollution with industrial, toxic, household, construction, and other waste	Bird conservation	PA Madzharovo
Prohibition the disturbance of the natural state of the water areas, water streams, their beds, banks and adjacent territories;	Bird conservation	PA Byala Reka
Prohibition the construction activities, forest felling, grazing and destruction of the undergrowth within radius of 300 m around nest of birds of prey and Black stork	Bird conservation	PA Byala Reka
Prohibition the construction activities, forest felling, grazing and destruction of the undergrowth within radius of 300 m around nest of birds of prey and Black stork	Bird conservation	PA Arda bridge
Species protection	Bats conservation	Whole ER
Excluding the possibility of legal application of poisons	Wolf conservation	Whole ER
Changes in conservation legislation	Biodiversity conservation	Whole ER
Prohibition of drainage activities and the like in the wetlands and their catchments	Bird conservation	PA Krumovitza
Prohibition the disturbance of the natural state of the water areas, water streams, their beds, banks and adjacent territories;	Bird conservation	PA Krumovitza
Prohibition the disposal of toxic waste	Bird conservation	PA Studen Kladenetz
Prohibition the establishment of new and enlargement of existing quarries and mines, destroying rocks	Bird conservation	PA Madzharovo
Prohibition the extraction of inert materials from the rivers beds	Bird conservation	PA Madzharovo
Prohibition the disposal of toxic waste	Bird conservation	PA Madzharovo
Prohibition the traffic of vehicles (including organization of motor and automobile competitions, outside the existing road network, forest and agriculture roads)	Bird conservation	PA Madzharovo
Making failsafe the electricity transmission network	Bird conservation	PA Madzharovo
Prohibition the construction and enlargement of infrastructure and facilities (including wind farms and related electricity transmission networks, sport centres, holyday villages, golf courses, etc.)	Bird conservation	PA Byala Reka
Prohibition the pollution with industrial, toxic, household, construction, and other waste	Bird conservation	PA Byala Reka
Prohibition the plugging, afforestation and change in the hydroelectric facilities (including mini hydroelectronic power-stations), as well as construction of dams, shoots and fishponds along the rivers	Bird conservation	PA Arda bridge
Prohibition the pollution with industrial, toxic, household, construction, and other waste	Bird conservation	PA Arda bridge
Administration and Regulation		
Declare the ER a Nature Park (IUCN V Category)	Protection of birds	Whole ER
Make a new PA (Luda Reka)	Protection of herpetofauna	Luda Reka
Construction of new power lines should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Improving the control on poisoning of wildlife	Bird conservation	Maglenik ridge
Management and control on newly determined for preservation natural territories at the conditions of various types of ownership (state, private and communal)	Forest conservation	Whole ER
Make several regions under protection (the hill Grebena / Sarta; Iranov hill / Iran tepe; the south-west outskirts of the town Ivailovgrad and the Ladja quarter; the outskirts of the village Mandritza and upwards the flow of the river Byala reka).	Insects conservation	Whole ER
Strict control	Reptiles conservation	Whole ER

Stop collection of turtles	Imperial Eagle conservation	Whole ER
Conservation of habitats	Biodiversity conservation	Whole ER
Enlargement of the PA Defileto	Protection of herpetofauna	Defileto
Control reinforcement (from Regional Inspection of Environment and Waters in Haskovo and Forest Service in Krumovgrad)	Conservation of the herpetofauna	Maglenik ridge
Alterations in the river beads should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Mines and geological activities should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Do not allow sand and stones gathering from river beds	Bird conservation	Maglenik ridge
Expand the borders of existing PAs	Herpetofauna conservation	Whole ER
Limited access of people	PAs management	PAs
To implement a strict regime and control of the use of the water resources	Plants and fungi conservation	Whole ER
Active participation in the establishment and registration of new protected natural territories and projects for expansion of the existing ones	Forest conservation	Whole ER
Elaboration and execution of regional and international programs for preventing erosion, fires, diseases and pests	Forest conservation	Whole ER
The practice of extracting of inert materials should be limited or stopped in the sections where the fish breed and the young fish concentrate.	Fish conservation	Whole ER
Strengthening the control measures for its fulfillment	Wolf conservation	Whole ER
Construction of new roads (except road repairing) should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Do not allow wind parks construction	Bird conservation	Maglenik ridge
Do not allow reconstructions	Bird conservation	Maglenik ridge
Do not allow mine activities	Bird conservation	Maglenik ridge
To announce a PA	Herpetofauna conservation	Whole ER
Do not disturb wildlife	PAs management	PAs
Do not allow stone-pits constructions	PAs management	PAs
Forbid mine activities	PAs management	PAs
Forbid fire making into the caves	PAs management	PAs
Urgent conservation of the locality of rare species in the region of v. Oreshari by establishment of a PA "Karst of Oreshari"	Plants and fungi conservation	Whole ER
Future co-operation with the mine enterprises' authorities on the field of the environment protection	Plants and fungi conservation	Whole ER
Management and control of protected natural territories in the forestry fund	Forest conservation	Whole ER
Expansion of the co-operation between the forestry projects, NGOs and movements and the local population for preservation and management of the protected natural territories and endangered areas	Forest conservation	Whole ER
Stopping the pollution of the dams with refuse waters and especially with waters containing heavy metal salts.	Fish conservation	Whole ER
Place large areas under protection	Reptiles conservation	Whole ER
Announcement of PAs	Imperial Eagle conservation	Whole ER
Forbid reconstructions in the area	Conservation of the herpetofauna	Maglenik ridge
Reconstructions and other activities leading to alteration of the landscape should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Construction of new and enlargement of existing stone-pits should not be allowed in the PA	Biodiversity conservation	Maglenik ridge

Construction of wind-power, water-power and sun-power farms should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Avoid plant damaging	PAs management	PAs
Reconstructions	PAs management	PAs
Do not allow geological studies	PAs management	PAs
Do not allow people to enter into the caves with bats	PAs management	PAs
A new PA to be established in the lower course of Mareshnitza river	Plants and fungi conservation	Whole ER
The boundaries of the PAs of "Patronka", "Momina skala" and "Meden kamak" have to be marked and appropriate information plates should be installed	Plants and fungi conservation	Whole ER
Participation in national, regional and international organizations, projects and programs on the problems of protected natural territories and projects and the environmental functions on the forestry projects	Forest conservation	Whole ER
Protection of the water reservoirs	Dragonflies conservation	Whole ER
Protection the vegetation around the water reservoirs	Dragonflies conservation	Whole ER
Enlargement of the protected territories /reservations/	Butterflies conservation	Whole ER
To submit proposals for legal protection and to develop management plans for the following bat roostings/territories: The cave Jarasa Ini and part of the rocky cliffs around; the karstic gorge Talaschmandere with situated on its slopes caves Samara, Prilepnata, Braschljanovata and Aina Ini; The rocky cliffs and the caves on the two riverbanks of Arda in the region of "Zeleznija Most"	Bats conservation	Whole ER
Creation of new PAs of different regime and widening of reserves	Conservation of large mammals	Whole ER
Creation of a PA	Conservation of the herpetofauna	Maglenik ridge
Limited access of vehicles	PAs management	PAs
Forbid climbing and other alternative sports	PAs management	PAs
To control the pollution	PAs management	PAs
To establish a natural reserve "The meanders of Byala reka"	Plants and fungi conservation	Whole ER
To forbid any construction activities, including building of roads, quarries, etc.	Plants and fungi conservation	Whole ER
To form a team of specialists for urgent intervention in case of anti-ecological acts directly or indirectly related to caves and the underground ambiance in general.	Cave fauna conservation	Whole ER
Announcement of PAs	Fish conservation	Whole ER
The practice of correcting the river beds should be stopped	Fish conservation	Whole ER
Protection of larger areas along the course of the river Byala reka	Reptiles conservation	Whole ER
Announcement of PAs	Bird conservation	Whole ER
Development of detailed management plans for the already protected roosts	Bats conservation	Whole ER
Research		***Whole ER***
Conduct more detailed survey on the dragonflies in the area	Dragonflies conservation	Whole ER
Study trophic relations within fish populations	Fish conservation	Whole ER
Determination of the species, needing urgent protection measures	Small mammals conservation	Whole ER
Morphological and biological Research on the species	Wolf conservation	Whole ER
Establish database of species distribution in the region that could be used for the aims of comparative studies and long-term-monitoring	Conservation of the small mammals (Micromammalia)	Whole ER
Elaborate a Red List for the invertebrate animals in the ER	Insects conservation	Whole ER
Study dynamics of fish populations	Fish conservation	Whole ER
Future Research	Small mammals conservation	Whole ER
The study of the nutrition of birds of prey	Small mammals conservation	Whole ER
Future Research on wolf population	Wolf conservation	Whole ER

Identification of the nutrition spectrum of the population	Wolf conservation	Whole ER
Elaborating species action plans	Biodiversity conservation	Whole ER
Identification of sites important for biodiversity	Biodiversity conservation	Whole ER
Survey on groups of plants and animals: almost all groups of insects and invertebrates; all plant and animal species that are subject to commercial use by man (herbs, snails, game animals, honey bees, fish, etc.)	Biodiversity conservation	Whole ER
Setting up of an Eastern Rhodope biodiversity database and using of the geographical information system (GIS) for tracking out changes	Biodiversity conservation	Whole ER
Investments for functional and comparative investigations on grazing impact on the biodiversity and structure of some biotopes (like pastures, shrubs), as well as on particular species (like Steffanofia daucoides)	Plants and fungi conservation	Whole ER
To establish scientific program for long-term monitoring	Plants and fungi conservation	Whole ER
Survey on the distribution and ecology of the mouse-tailed dormouse and the grey hamster	Small mammals conservation	Whole ER
To work on a long-term monitoring programme to survey conditions of the caves and their fauna.	Cave fauna conservation	Whole ER
Continuation of the study of the damages, caused by the wolf	Wolf conservation	Whole ER
To make specific management plans for the PAs	Biodiversity conservation	Whole ER
Study the opportunities for economic development of the region	Biodiversity conservation	Whole ER
Further studies on the status and ecology of spiders in E Rhodopes	Spiders conservation	Whole ER
Perform a study of the invertebrate fauna in the reserves	Insects conservation	Whole ER
To undertake complete Research of the fauna in the ER	Insects conservation	Whole ER
Study the sources of dam pollution	Fish conservation	Whole ER
Study fish migration	Fish conservation	Whole ER
Study flow connection between standing dams and the transfer of biological materials	Fish conservation	Whole ER
Research on the number and density of the species	Wolf conservation	Whole ER
Proposing priority sites for protection	Biodiversity conservation	Whole ER
To collect additional information on the poorly studied areas: Kurdzhli reservoir; the area east of river Borovitza; the area around the highest peak of the mountain (Veikata); the area from the village of Chakalarovo west to the saddleback Tri Kamaka as well as the border area with Greece from the village of Veikata east to the peak Ushite	Biodiversity conservation	Whole ER
Research on ecosystems productivity	Biodiversity conservation	Whole ER
Monitoring		***Whole ER***
Monitoring of the large bat colonies in the region	Bats conservation	Whole ER
Monitoring of habitats	Biodiversity conservation	Whole ER
Monitoring of the qualifying species and their habitats	Bird conservation	PA Krumovitza
Monitoring of the qualifying species and their habitats	Bird conservation	PA Byala Reka
Monitoring the effectiveness of the conservation efforts	Bird conservation	PA Byala Reka
Monitoring	Biodiversity conservation	Whole ER
Monitoring of the qualifying species and their habitats	Bird conservation	PA Studen Kladenetz
To undertake regular Monitoring	Fish conservation	Whole ER
Monitoring of the souslik's population	Small mammals conservation	Whole ER
Monitoring of sites and territories	Biodiversity conservation	Whole ER
Monitoring the effectiveness of the conservation efforts	Bird conservation	PA Studen Kladenetz
Monitoring the effectiveness of the conservation efforts	Bird conservation	PA

		Madzharovo
Monitoring of the qualifying species and their habitats	Bird conservation	PA Arda bridge
Monitoring of the qualifying species and their habitats	Bird conservation	PA Madzharovo
Monitoring the effectiveness of the conservation efforts	Bird conservation	PA Arda bridge
Conduct Monitoring surveys of bioindicator small mammal species	Conservation of the small mammals (Micromammalia)	Whole ER
Monitoring of species, populations and breeds	Biodiversity conservation	Whole ER
Monitoring of threats	Biodiversity conservation	Whole ER
Landscape conservation		
Such zones of concrete regime of protection must be envisaged in areas identified as potential natural "ecological corridors" of dispersal for the brown bear and the chamois	Conservation of large mammals	Whole ER
Prohibition of drainage activities and the like in the wetlands and their catchments	Bird conservation	PA Madzharovo
Prohibition the removal of specific landscape features	Bird conservation	PA Madzharovo
Prohibition the changes of designation and use of meadows, pastures, glades, marches, wetlands and water streams	Bird conservation	PA Byala Reka
Prohibition the removal of specific landscape features	Bird conservation	PA Byala Reka
Keep some humid areas within forests	Bats conservation	Maglenik ridge
Prohibition the changes of designation and use of meadows, pastures, glades, marches, wetlands and water streams	Bird conservation	PA Krumovitza
Prohibition the drainage activities and their like in wetlands and their catchments	Bird conservation	PA Studen Kladenetz
Prohibition the removal of specific landscape features	Bird conservation	PA Arda bridge
Conservation of natural riversides (otter); wooded valleys, forests - zones of permanent wolf lairs; mixed pastoral landscapes and low mountain crossed and desolate rocky areas (marbled polecat and jackal)	Conservation of large mammals	Whole ER
Prohibition the removal of specific landscape features	Bird conservation	PA Krumovitza
Prohibition the changes of designation and use of meadows, pastures, glades, marches, wetlands and water streams	Bird conservation	PA Studen Kladenetz
Prohibition the changes of designation and use of meadows, pastures, glades, marches, wetlands and water streams	Bird conservation	PA Madzharovo
Prohibition the changes of designation and use of meadows, pastures, glades, marches, wetlands and water streams	Bird conservation	PA Arda bridge
Agriculture and livestock rearing		
Strict ban on the usage of pesticides in forests and non-cultivated lands	Lepidoptera group and their habitats conservation	Whole ER
Strict ban on firing of stubble fields	Lepidoptera group and their habitats conservation	Whole ER
Grazing in the forests should not be allowed in the PA (PA)	Biodiversity conservation	Maglenik ridge
Changes in the permanent use of pastures and meadows should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Do not allow goat grazing within forests	Bird conservation	Maglenik ridge
Restriction or prohibition of the livestock grazing on small territories with natural and easy for pointing borders	Insects conservation	Whole ER
Restriction of goats grazing in small territories and on territories with predominant afforesting with oak or other trees from the belt of the oak	Insects conservation	Whole ER
Ban of pastures in some areas	Butterflies conservation	Whole ER

Ban and control over the burning of stubble fields	Butterflies conservation	Whole ER
Prohibition the growing of genetically modified plants	Bird conservation	PA Krumovitza
Preventive measures for the protection of agriculture, stock-farming and forestry from predators	Conservation of large mammals	Whole ER
Avoid overgrazing and under grazing	Plants conservation	Maglenik ridge
Maintenance of pastures through moderate grazing by livestock and/or prevention of shrub succession	Bird conservation	PA Madzharovo
Maintenance of the traditional crops and the mosaic character and diversity of the arable lands and crops	Bird conservation	PA Madzharovo
Maintenance of the traditional crops and the mosaic character and diversity of the arable lands and crops	Bird conservation	PA Arda bridge
Control on mowing and pasture in natural habitats and establishing new PAs	Lepidoptera group and their habitats conservation	Whole ER
Forbid the use of pesticides	Bats conservation	Maglenik ridge
Do not allow changes in permanent pastures' use	Bird conservation	Maglenik ridge
Grazing control	PAs management	PAs
Total ban on mowing of natural vegetation in some areas	Butterflies conservation	Whole ER
Prohibition the use of artificial fertilizers and other chemical substances, poisons and other nonselective instruments for combating pests	Bird conservation	PA Krumovitza
Maintenance of the traditional crops and the mosaic character and diversity of the arable lands and crops	Bird conservation	PA Krumovitza
Maintenance of the pastures through moderate grazing by livestock and/or prevention of shrub succession	Bird conservation	PA Studen Kladenetz
Prohibition the growing of genetically modified plants	Bird conservation	PA Byala Reka
Maintenance of the traditional crops and the mosaic character and diversity of the arable lands and crops	Bird conservation	PA Byala Reka
Promote extensive grazing in the area	Bird conservation	Maglenik ridge
Forbid pesticides use	PAs management	PAs
Total ban on the use of pesticides in the forest and the wild vegetation areas in E Rhodopes	Butterflies conservation	Whole ER
Prohibition the use of artificial fertilizers and other chemical substances, poisons and other nonselective instruments for combating pests	Bird conservation	PA Studen Kladenetz
Prohibition the growing of genetically modified plants	Bird conservation	PA Studen Kladenetz
Maintenance of the traditional crops and the mosaic character and diversity of the arable lands and crops	Bird conservation	PA Studen Kladenetz
Prohibition of haymaking and harvesting of crops in the localities of the Montagu's Harrier before 1 August	Bird conservation	PA Byala Reka
Maintenance of pastures through moderate grazing by livestock and/or prevention of shrub succession	Bird conservation	PA Arda bridge
Mowing after 15 of June	Bird conservation	Maglenik ridge
To set a minimum limits of the grazing	Plants and fungi conservation	Whole ER
Prohibition the use of artificial fertilizers and other chemical substances, poisons and other nonselective instruments for combating pests	Bird conservation	PA Madzharovo
Maintenance of pastures through moderate grazing by livestock and/or prevention of shrub succession	Bird conservation	PA Byala Reka
Prohibition the creation of monoculture field blocks larger than 1 ha	Bird conservation	PA Arda bridge
Forest management		
Intensity of the tree fall of more than 20% and repeatability	Biodiversity conservation	Maglenik

under 7-9 yrs in the same area should not be allowed in the PA		ridge
Mark the trees that are known being used by bats	Bats conservation	Maglenik ridge
Maintenance of subcanopy cover at about 0.3	Bats conservation	Maglenik ridge
Protect old trees and trees with hollows	Bird conservation	Maglenik ridge
Stop all forestry activities within 500 m radius around birds of prey or black stork nests	Bird conservation	Maglenik ridge
Complex approach in forest management and biodiversity conservation	Sustainable use of forests	Maglenik ridge
Close-to-nature forestry practices	Sustainable use of forests	Maglenik ridge
Keep some amount of dead wood within forests	Sustainable use of forests	Maglenik ridge
To sustainably manage forests, a close cooperation with the local Forestry authorities is recommended	Plants and fungi conservation	Whole ER
Information campaign related to sustainable forest management should be made for the forestry staff	Plants and fungi conservation	Whole ER
Separation of seed bases from natural species of oaks and especially beech trees	Forest conservation	Whole ER
The reconstruction rates of forests must be determined carefully, since the distribution of the categories by age in the E Rhodopes is unfavorable	Forest conservation	Whole ER
Increasing the proportion of the forestry activities as a whole on the basis of rational utilization of the forest resources with correct care and protection of the same	Forest conservation	Whole ER
Alternative sources of heating must be found (the wood being the only source of heating for the population of the region).	Butterflies conservation	Whole ER
Prohibition the clearfells	Bird conservation	PA Madzharovo
Prohibition the felling of shrub-like forests and shrubs on certain places;	Bird conservation	PA Madzharovo
Prohibition the clearfells	Bird conservation	PA Byala Reka
Prohibition the felling on terrains on slopes > 25 degrees	Bird conservation	PA Byala Reka
Prohibition the felling of shrub-like forests and shrubs on certain places;	Bird conservation	PA Byala Reka
Cease tree extraction in the preserved old-growth forests	Bird conservation	PA Byala Reka
Increase the periods between the fellings, clearfells, retention of hollow trees	Bird conservation	PA Byala Reka
Prohibition riverside forests felling 30 to 50 m from the river bank	Bird conservation	PA Arda bridge
Restoration of the natural forest and river vegetation	Bird conservation	PA Arda bridge
Forbid clear felling everywhere in the Easter Rhodopes	Lepidoptera group and their habitats conservation	Whole ER
In further afforestation only local tree species must be used	Lepidoptera group and their habitats conservation	Whole ER
Planting only local broadleaved tree species	Bats conservation	Maglenik ridge
Protect the trees with nests of birds of prey and black storks	Bird conservation	Maglenik ridge
Keep at min 20 cubic metres lying dead wood per ha	Sustainable use of forests	Maglenik ridge
When fallen by natural disturbances, trees should not be	Sustainable use of forests	Maglenik

removed from the forest		ridge
Forbid clear cut	PAs management	PAs
A long-term strategy for species conservation requires an adequate management of the forests within the localities, as well as in the corresponding basins.	Plants and fungi conservation	Whole ER
Protection of old trees by the law as a natural sight	Plants and fungi conservation	Whole ER
To inform the local Forestry about the problem with cutting old and threatened trees and to work towards a change of the forestry-management plan (if needed)	Plants and fungi conservation	Whole ER
Stopping the planting with coniferous arboricultures, except of the places where a recultivation of the terrain took place	Plants and fungi conservation	Whole ER
Cease tree extraction in the preserved old-growth forests	Bird conservation	PA Krumovitza
Cease tree extraction in the preserved old-growth forests	Bird conservation	PA Studen Kladenetz
Prohibition the felling of hollow and dying trees in the breeding areas of hole-nesting birds or carrying out big scale sanitary fellings	Bird conservation	PA Madzharovo
Prohibition the felling on terrains on slopes > 25 degrees	Bird conservation	PA Madzharovo
Increase the periods between the fellings, clearfells, retention of hollow trees	Bird conservation	PA Madzharovo
Prohibition the clearfells	Bird conservation	PA Arda bridge
Increase the periods between the fellings, clearfells, retention of hollow trees	Bird conservation	PA Arda bridge
Tree fell during the period April-June should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Use of heavy machines for wood transportation should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Tree fall of trees with hollows and DBH > 30 cm should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Make network of trees (25-30 ha) with hollows, loose bark and dead wood	Bats conservation	Maglenik ridge
Make inforest corridors/openings	Bats conservation	Maglenik ridge
In above 50 yrs old forests, keep 20-25 standing dead or dying trees (with DBH >22 cm, if possible) per ha	Sustainable use of forests	Maglenik ridge
Avoid grouping of the remnants from the tree fall (branches, trunks etc.)	Sustainable use of forests	Maglenik ridge
Protect trees that are proved as used by some threatened species	Sustainable use of forests	Maglenik ridge
Better control of forestry practices	PAs management	PAs
No more clear cuts and reconstructions	Plants conservation	Maglenik ridge
Step-by-step replacing of the coniferous cultures with local deciduous species	Plants and fungi conservation	Whole ER
Further transformation of the offshoot forests into seed forests by means of various methods.	Forest conservation	Whole ER
Sawing to be allowed only for dead trees or atypical and artificial introduced ones as pseudoacacia and artificial pine trees.	Butterflies conservation	Whole ER
In afforestation only local forest types to be planted only in locations where they used to grow.	Butterflies conservation	Whole ER
Prohibition the felling of shrub-like forests and shrubs on certain places;	Bird conservation	PA Krumovitza
Increase the periods between the fellings, clearfells, retention of hollow trees	Bird conservation	PA Krumovitza
Increase the periods between the fellings, clearfells, retention of hollow trees	Bird conservation	PA Studen Kladenetz

Replacement of the artificial cultures with local species	Bird conservation	PA Arda bridge
In felling, only dead trees are to be cut and introduced foreign tree species	Lepidoptera group and their habitats conservation	Whole ER
Protection of the neighbour to burned areas for the restoration of the affected populations	Conservation of tortoises	Whole ER
Forbid the dead wood removal from forests	Conservation of the herpetofauna	Maglenik ridge
Increase the period for forest rotations	Conservation of the herpetofauna	Maglenik ridge
Tree fell in oak-dominant, beech-dominant and mixed forests should not be allowed in the PA (except if this is not allowed by the PA 's management plan)	Biodiversity conservation	Maglenik ridge
Introducing non-local plant and animal species should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Old forests (>100 yrs old) protection	Bats conservation	Maglenik ridge
Protection of old trees and providing conditions for more sun radiation around them to benefit higher abundance of insects	Bats conservation	Maglenik ridge
Partial opening of the canopy cover to insure more sunlight penetration	Bats conservation	Maglenik ridge
Make outer forest edges with depth no less than 30 m	Bats conservation	Maglenik ridge
Stop forestry activities until the official declaration of the PA	Bird conservation	Maglenik ridge
Keep the dead wood in min 10% of standing wood	Sustainable use of forests	Maglenik ridge
Ensure forest reservoirs (at least 5% of the territory should be maintained as forests in good conditions)	Sustainable use of forests	Maglenik ridge
Trees with symptoms of dying and without any economical value should not be removed from the forest	Sustainable use of forests	Maglenik ridge
Main forestry activities must guarantee their wholesome development, provide for their optimum utilization for the production of timber for the regional and national economies, taking into account the biological characteristics of the individual tree species	Forest conservation	Whole ER
Reinforcement of the structures for guarding and protection of the forests and creation of safe labor conditions in the forestry projects	Forest conservation	Whole ER
Prohibition the felling of hollow and dying trees in the breeding areas of hole-nesting birds or carrying out big scale sanitary fellings	Bird conservation	PA Krumovitza
Cease tree extraction in the preserved old-growth forests	Bird conservation	PA Madzharovo
Prohibition the felling of hollow and dying trees in the breeding areas of hole-nesting birds or carrying out big scale sanitary fellings	Bird conservation	PA Byala Reka
Forest rotation period should be prolonged in the PA	Biodiversity conservation	Maglenik ridge
Reconstructions in deciduous and mixed forests should not be allowed in the PA	Biodiversity conservation	Maglenik ridge
Do not allow tree fall in > 80 yrs old forests	Bird conservation	Maglenik ridge
Do not allow tree fall along the rivers	Bird conservation	Maglenik ridge
A limit for the cutting of beech forests in the region of Bubino should be requested in respect that only sanitary felling should be performed in the future.	Plants and fungi conservation	Whole ER
Proper localization of old trees on the forestry-management	Plants and fungi conservation	Whole ER

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To inform local authorities (administrative and border) about the problem with cutting old and threatened trees	Plants and fungi conservation	Whole ER
Aid in favour of the initiatives of the municipalities, legal entities and persons, NGOs and NGAs in the care of forests and their multi-functional utilization in accordance with the Law about Forests	Forest conservation	Whole ER
Total prohibition of forest cutting on the territory of ER (or to be allowed only in extreme cases under tight control and no area to be totally cut).	Butterflies conservation	Whole ER
Restoration of the natural forest and river vegetation	Bird conservation	PA Krumovitza
Replacement of the artificial cultures with local species	Bird conservation	PA Krumovitza
Prohibition the clearfell	Bird conservation	PA Studen Kladenetz
Restoration of the natural forest and river vegetation	Bird conservation	PA Studen Kladenetz
Replacement of the artificial tree plantations with local species	Bird conservation	PA Studen Kladenetz
Prohibition the felling of hollow and dying trees in the breeding areas of hole-nesting birds or carrying out big scale sanitary fellings	Bird conservation	PA Arda bridge
Prohibition the felling of shrub-like forests and shrubs on certain places;	Bird conservation	PA Arda bridge
Cease tree extraction in the preserved old-growth forests	Bird conservation	PA Arda bridge
Wildlife management		
Implementation of management plans	Conservation of the herpetofauna	Whole ER
The breeding of the Pike-perch and the Wels in the dams should be supported by setting up special nests at suitable places.	Fish conservation	Whole ER
Artificial insemination	Lesser Kestrel conservation	Whole ER
Reintroduction	Lesser Kestrel conservation	Whole ER
Prohibition the construction activities, forest felling, grazing and destruction of the undergrowth within radius of 300 m around nest of birds of prey and Black stork	Bird conservation	PA Krumovitza
Prohibition the construction activities, forest felling, grazing and destruction of the undergrowth within radius of 300 m around nest of birds of prey and Black stork	Bird conservation	PA Studen Kladenetz
Prohibition the access of people and human activities within a radius of 300 to 500 m around nests of birds of prey and Black Stork in the breeding season (March-July)	Bird conservation	PA Madzharovo
Establish the population density of food source necessary to sustain the imperial eagles nesting in the region	Conservation of the small mammals (Micromammalia)	Whole ER
Protect the habitats and ecological corridors between the subpopulations of the wolf, jackal and marbled polecat	Conservation of large mammals	Whole ER
Make tables to point the attention on the wildlife crossing over the roads	Conservation of the herpetofauna	Maglenik ridge
Support the populations of some fish species	Fish conservation	Whole ER
The carp population should be maintained by means of stocking based on scientifically grounded norms of the quantities and size of the stock material.	Fish conservation	Whole ER
Provision of a nourishment basis	Imperial Eagle conservation	Whole ER
Establishment of animal passes during the renovation of existing roads and the construction of new ones	Bird conservation	PA Krumovitza
Prohibition of rock climbing in the period February - August except on places unsuitable for nesting of birds of prey	Bird conservation	PA Studen Kladenetz
Establishment of animal passes during the renovation of existing roads and the construction of new ones	Bird conservation	PA Byala Reka

Prohibition of rock climbing in the period February - August except on places unsuitable for nesting of birds of prey	Bird conservation	PA Arda bridge
Prohibition all activities that degrade the quality of the habitats or cause disturbance to birds (except on designated places and under conditions in accordance with a management plan)	Bird conservation	PA Arda bridge
Implementation of species action plans	Conservation of the herpetofauna	Whole ER
Making failsafe the electricity transmission network	Bird conservation	PA Arda bridge
Solving the problem with feral dogs (and cats?)	Conservation of large mammals	Whole ER
The preservation of the threatened populations of insects is more practical to be performed through the preservation of their places of inhabitation, instead through the implementation of measures for the single species.	Insects conservation	Whole ER
There are six important habitats for preservation of the discussed groups of insects (xerothermic herbaceous vegetation; mesophytic herbaceous vegetation without the bracken; eternal green xerophyte bushes of Mediterranean type; xero- and mesophyte bushes with falling leaves, without lilac; xerophyte oak-tree forests; sands nearby rivers).	Insects conservation	Whole ER
Replacement of the mercury luminescent street in the towns and villages in the region that attract a lot of night insects with red spectrum lamps which does not attract so much insects.	Butterflies conservation	Whole ER
Building of new roosts - as bat boxes etc.	Bats conservation	Whole ER
Prohibition of rock climbing in the period February - August except on places unsuitable for nesting of birds of prey	Bird conservation	PA Krumovitza
Prohibition the access of people and human activities within a radius of 300 to 500 m around nests of birds of prey and Black Stork in the breeding season (March-July)	Bird conservation	PA Studen Kladenetz
Prohibition the construction activities, forest felling, grazing and destruction of the undergrowth within radius of 300 m around nest of birds of prey and Black stork	Bird conservation	PA Madzharovo
Prohibition the access of people and human activities within a radius of 300 to 500 m around nests of birds of prey and Black Stork in the breeding season (March-July)	Bird conservation	PA Byala Reka
Prohibition the access of people and human activities within a radius of 300 to 500 m around nests of birds of prey and Black Stork in the breeding season (March-July)	Bird conservation	PA Arda bridge
To undertake restoration activities	Fish conservation	Whole ER
To mitigate the human-predator conflict by creation of population management plans for problematic carnivore species of conservation value	Conservation of large mammals	Whole ER
Artificial feeding of birds of prey	Bird conservation	Maglenik ridge
Protect the caves, where Centromerus milleri inhabits	Protect Centromerus milleri	Whole ER
Prohibition the access of people and human activities within a radius of 300 to 500 m around nests of birds of prey and Black Stork in the breeding season (March-July)	Bird conservation	PA Krumovitza
Hunting & fishing		
Fight against the poaching especially in the breeding season	Fish conservation	Whole ER
Fight against the poaching especially around migration routes of fish	Fish conservation	Whole ER
If industrial fishing is to be organized in the dams the allowed species and quantities should be determined in compliance with the condition of the populations on the basis of preliminary scientific research.	Fish conservation	Whole ER
Prohibition the commercial and recreational fishing in the breeding and/or congregation areas of waterfowl	Bird conservation	PA Studen Kladenetz

Forbid hunting and fishing	PAs management	PAs
Measures should be undertaken against the fishing with net devices, electric power, explosives and poisoning substances.	Fish conservation	Whole ER
Preservation of Hunting Grounds	Imperial Eagle conservation	Whole ER
Prohibition the game breeding and hunting except on places and under conditions defined by a management plan	Bird conservation	PA Krumovitza
Improvement the organization of warding and fighting the poaching	Fish conservation	Whole ER
In the national parks and the reserves hunting the wolf must be strictly prohibited	Wolf conservation	Whole ER
Prohibition the game breeding and hunting except on places and under conditions defined by a management plan	Bird conservation	PA Studen Kladenetz
Prohibition the game breeding and hunting except on places and under conditions defined by a management plan	Bird conservation	PA Byala Reka
Prohibition the game breeding and hunting except on places and under conditions defined by a management plan	Bird conservation	PA Arda bridge
Fight against the poaching especially in the places where the fish spawn	Fish conservation	Whole ER
Do now allow hunting during the birds of prey breeding season (February - August)	Bird conservation	Maglenik ridge
Protect fish diversity in the conditions of growing fishing pressure by sport fishing and recreation tourism.	Fish conservation	Whole ER
Restrict hunting of rabbits	Imperial Eagle conservation	Whole ER
Placing of poisoned food and traps against predators must not be admissible	Imperial Eagle conservation	Whole ER
Tourism & environmental education		
Increasing the ecological knowledge of local people (including foresters)	Bird conservation	Maglenik ridge
To educate the local people	Plants and fungi conservation	Whole ER
To publish information materials	Plants and fungi conservation	Whole ER
Education activities	Reptiles conservation	Whole ER
To organize educational workshops for local control authorities	Plants and fungi conservation	Whole ER
Defining and marking tourist areas	Bird conservation	PA Arda bridge
Developing eco- or environment friendly tourism	Development of tourism	Whole ER
Social activities	Imperial Eagle conservation	Whole ER
Education program with local environmental authorities and schools	Bats conservation	Whole ER
Drawing-up an educational programme, which should be in accordance with the species condition in the region and the psychology of the local population	Wolf conservation	Whole ER
Defining and marking of the tourist areas	Bird conservation	PA Krumovitza
Defining and marking tourist areas	Bird conservation	PA Studen Kladenetz
Creation of conditions stimulating ecotourism, in places - game tourism	Conservation of large mammals	Whole ER
Starting an educational and information campaign	Conservation of the herpetofauna	Maglenik ridge
Implementing an educational and informational campaign in the region for popularizing the conservation status of the species and necessity to protect them	Herpetofauna conservation	Whole ER
To install information plates for indication of reserve boundaries, the management regime of water resources, forbid grazing, forbid other human activities endangering the nature, forbid activities causing direct and indirect pollution, indication of the tourist routes, forbid leaving the tourist routes, install plates with information on the	Plants and fungi conservation	Whole ER

biodiversity in the area		
Defining and marking tourist areas	Bird conservation	PA Madzharovo
Defining and marking tourist areas	Bird conservation	PA Byala Reka
To install information plates in the vicinity to the old trees - to inform about its conservation status and to forbid cutting and picking of branches.	Plants and fungi conservation	Whole ER
To publish educational materials on the caves and the life in them.	Cave fauna conservation	Whole ER
Live discussions	Imperial Eagle conservation	Whole ER
High-quality propaganda materials	Imperial Eagle conservation	Whole ER
Environmental education	Bats conservation	Whole ER
Community involvement	Biodiversity conservation	Whole ER
Sustainable development		
It is necessary to find an alternative source of heating	Lepidoptera group and their habitats conservation	Whole ER
Improve the nature-protective culture of local people, hunters and foresters	Bird conservation	Maglenik ridge
Implementation of modern technologies of higher ecological and ergonomical properties and use of technologies, which guarantee the maximum protection of the reproduction and existing plants	Forest conservation	Whole ER
Elaboration of regional strategy and plan for Sustainable development	Biodiversity conservation	Whole ER
Mercury-vapor lamps in settlements and on single buildings to be replaced by halogen lamps with reddish or orange light	Lepidoptera group and their habitats conservation	Whole ER